

Final Project

Kevin A. Rivera

Due Date : December 1st, 2025

EML 4024C - ENGINEERING DESIGN PRACTICE

Dr. Shad Abdel

Monday 11:30am - 02:30pm

I. Abstract

Why: Roadside infrastructure should be evaluated for safety and structural reliability; to do this it is important to understand how extreme weather events, like hurricane level winds, affect traffic stop signs via proper assessment. The 30 x 30 inch MUTCD standard stop sign, required on all public roadways, can experience high aerodynamic forces and destructive oscillation during severe storms. Smaller stop signs are present in residential and private areas, but the 30 inch sign is critical to analyze due the larger presence and high expected wind loading of stop signs used in areas of critical infrastructure.

What: This project aimed to analyze the structural response of the 30 x 30 inch stop sign under Category 2 hurricane conditions. The main focus of the study is on the behavior of the sign when under extreme aerodynamic loading.

How: A SolidWorks model of the aluminum plate (the stop sign) and the steel U channel beam for the post was created and attached to make a stop sign. Then it was evaluated using three analysis methods. The first being fluid dynamics analysis using Finite Volume analysis (FVA) was used to simulate airflow around the sign with speeds representing a category two hurricane; this simulation allows us to determine the pressure distribution and drag force on the sign. Then the structural model was put through a static structural analysis using Finite Element Analysis to quantify stresses, strains, and deformation under peak wind loading based off of the results of the drag force from the previous simulation. Finally, a dynamic structural analysis using FEA was performed to analyze how the sign may behave during the hurricane from the vibrations caused by the weather.

Important results: The flow simulation showed that the 30 × 30 inch stop sign generates a steady drag force of approximately 206.86 lbf under Category 2 hurricane wind speeds, with effectively zero lift, confirming that loading is dominated by frontal pressure. The static FEA revealed that the maximum von Mises stress of the assembly occurs at the fixed base of the steel post and reaches approximately 124,000 psi, exceeding yield strength and indicating structural failure under the simulated wind load. Maximum displacement at the top of the sign was approximately 5.7 inches, while strain was concentrated almost entirely at the lower third of the post. Modal analysis produced natural frequencies of 0.59 Hz, 6.96 Hz, and 48.83 Hz, showing that the assembly behaves as a flexible cantilever dominated by a low-frequency global bending mode.

Discussion of results: Overall, the results indicate that the stop-sign assembly is highly vulnerable to upper-range Category 2 hurricane winds. Aerodynamic loading is dominated by steady drag forces created by full flow separation around the flat plate, generating large bending moments at the post base. The FEA shows that the U channel post, not the aluminum plate, is the structural weak point, experiencing stresses far above its elastic limit and undergoing significant lateral deflection. The modal analysis supports this interpretation: the low fundamental frequency reflects a tall, flexible structure that can undergo large amplitudes swaying under turbulent wind. Higher order modes are too stiff and too high in frequency to be excited by hurricane scale flow.

Together, the simulations demonstrate that under extreme wind loading, a standard U channel post is insufficient to prevent failure, and reinforcement or redesign would be required for hurricane-prone environments.

II. Relevance

Practical Application: Stop signs are a critical traffic control device and their failure can create dangerous conditions. For example during evacuations or emergency response, a stop sign that has flown away or been destroyed would no longer help regulate traffic. The stop sign itself flying away is dangerous itself. Understanding the effects of high wind on the 30 x 30 inch MUTCD standard stop sign determines the typical structural integrity and if the normal installation can withstand extreme weather. The result of this project can guide future improvements in sign design.

III. Introduction

Hurricane force winds impose large amounts of stress on stop signs creating a harsh aerodynamic and structural environment. This project attempts to evaluate how a 30 x 30 inches responds to a category 2 hurricane. A 30 x 30 stop was chosen as the analysis subject due to the larger surface area and it being required on all public roads as a standard wet by the FDOT. Below is a table of sign size and placement. The analysis focused on how aerodynamic forces interact with the aluminum plate and steel post, and whether the sign experiences high stress, deformation, or vibration that could lead to failure.

Table 2.39-3. Stop and Stop Ahead Sign Sizes and Placement

POSTED SPEED (mph)	STOPPING SIGHT DISTANCE (feet)	STOP SIGN SIZE ¹ (inches)	STOP SIGN RECOGNITION DISTANCE (20/70) (feet)	STOP AHEAD SYMBOL SIGNS ² (inches)	STOP AHEAD SIGN PLACEMENT DISTANCE (feet)
20	150	24	178	—	—
25	200	30	222	—	—
30	250	36	267	36*	125*
35	300	36	267	36*	175*
45	450	36	267	36	325
50	550	48	356	36	425
55	625	48	356	36	500
*If needed for restricted sight distance locations in urban areas.					
¹ On state highways, the 48-inch STOP sign should be considered for 45 mph or greater. STOP signs on roads intersecting the state highway are usually replaced in FDOT construction projects. The sizes in this section are recommended for the replacement signs. Motorists traveling on local roads, in urban areas, expect to encounter STOP signs. STOP signs larger than 36-inches should be used when greater emphasis or visibility is needed.					
² On state highways, in rural areas, motorists may not expect to encounter a STOP sign. As an enhancement, the STOP AHEAD sign should be used for speeds equal to or greater than 45 mph. On local roads, in rural areas, motorists usually expect to stop as they cross a state highway. Where sight distance restrictions exist, a STOP AHEAD sign should be used.					

Figure 1. Stop Sign Sizes and Placements

Objectives: The primary objective of this project was to evaluate the aerodynamic and structural behavior of a standard 30 × 30 inch MUTCD stop sign when subjected to Category 2 hurricane wind loading. A full model of the octagonal aluminum plate and galvanized steel U channel post was created in SolidWorks to represent a typical roadside installation. A time dependent external flow simulation was performed in SolidWorks Flow Simulation to determine the pressure distribution, drag force, lift force, and wake characteristics generated by 110 mph hurricane winds. The computed aerodynamic drag force was then applied in a static Finite Element Analysis (FEA) study to quantify stress, strain, and displacement in the post–sign assembly under peak wind loading. A modal (vibration) FEA study was conducted to calculate the first three natural frequencies and mode shapes, assessing susceptibility to wind-induced oscillation or resonance. Simulation outputs were compared with simplified hand calculations using cantilever beam theory to verify displacement and bending behavior of the signpost under lateral loading.

Outline: The report begins with a description of the modeling approach of constructing the stop sign in solidworks and the simulation approach using FVA and FEA. The result section shows

the geometry of the stop sign and post and all associated plots and tables. This is then followed by a discussion which interprets all included results and a verification and reliability discussion. Finally the report concludes by summarizing the stop signs performance under hurricane loading and providing recommendations.

IV. Methodology

Simulations were run using a constructed model of a Stop sign to examine and analyze behavior in a category two hurricane. The unit system used for the following was inch-pound-second. A combined Finite Volume (FV) and Finite Element (FE) approach was used to capture both the aerodynamic and structural behavior of the 30 × 30 inch stop sign. The airflow was modeled with an FV solver using a $k-\omega$ SST turbulence model, chosen for its accuracy in predicting separated, high-speed flow and vortex shedding. The resulting pressure field was applied to the structural model, which was analyzed using static FEA to determine stresses and deformation under peak wind loads. A modal solver was then used to identify natural frequencies and mode shapes, followed by a transient dynamic solver to evaluate the sign's response to oscillatory, vortex-shedding-based loading. These solvers were selected to accurately represent both steady and dynamic effects of hurricane-force winds while maintaining computational efficiency.

Geometrical Modeling:

The first step was to create the stop sign model in Solidworks. The first component of the sign models was the octagonal plate. The plate followed these dimensions below with 5052 H38 alloy was the assigned material along with a red base plate color and white letter to imitate the real life version:

- 30 x 30 octagonal plate
- The mounting holes 24 inches apart

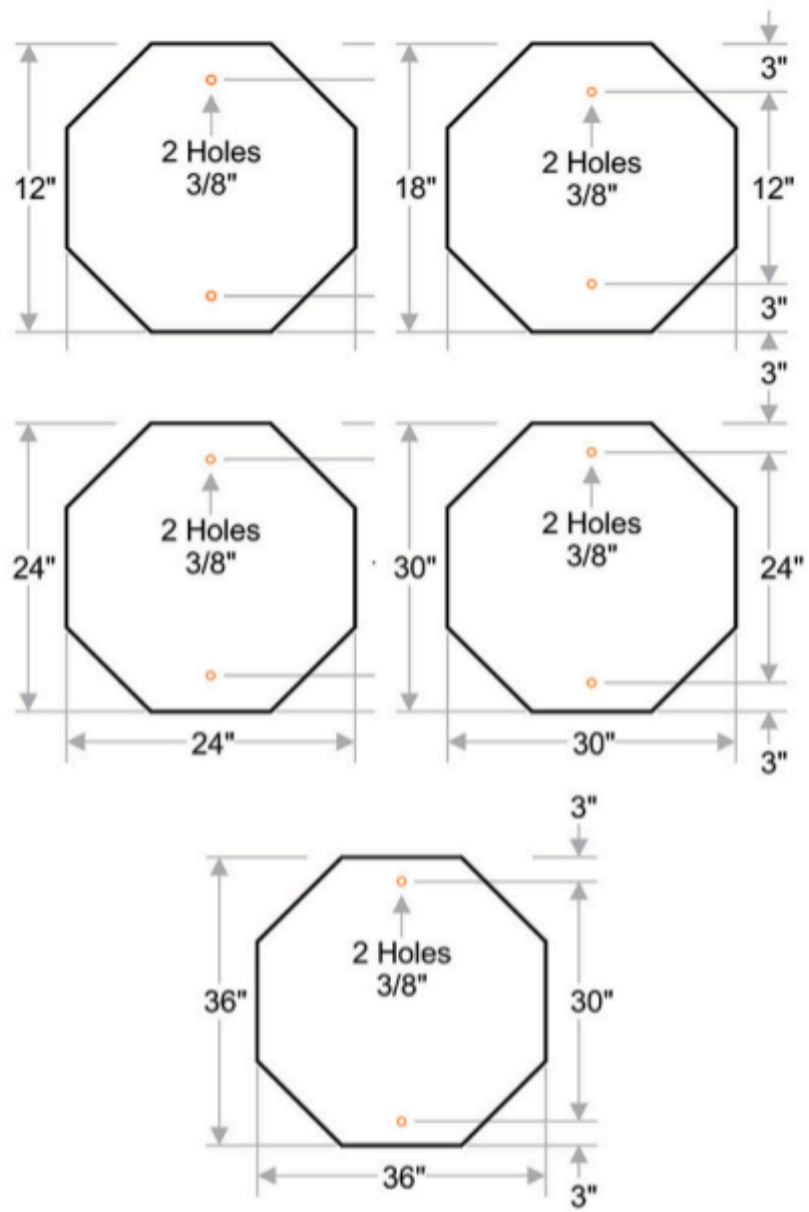


Figure 2. Reference dimensions for Stop Sign Plate

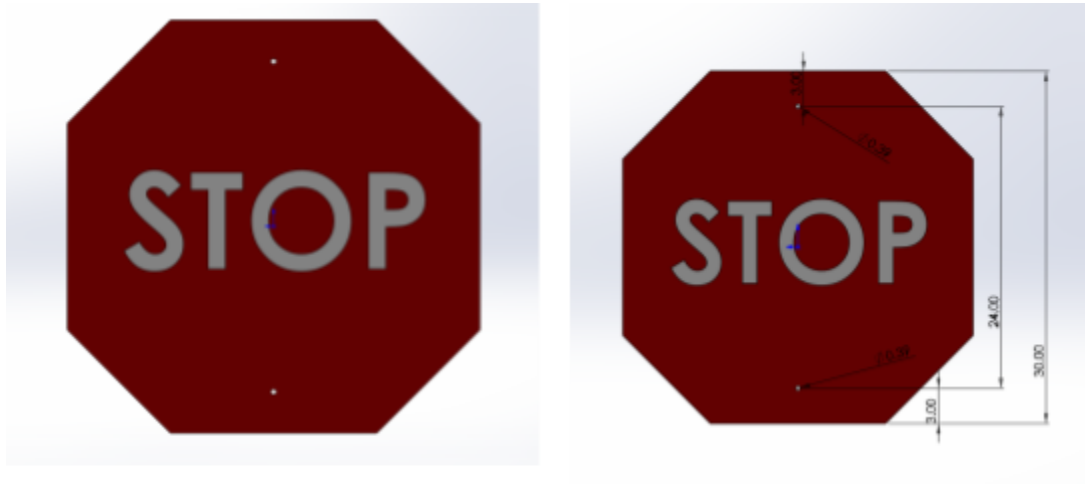


Figure 3. Solid Works Stop sign Without (left) with (right) dimesions

Material Properties - 5052 H38

Property	Value	Units
Elastic Modulus	70,000	N/m m ²
Poisson's Ratio	0.33	—
Shear Modulus	25,900	N/m m ²
Mass Density	2,680	kg/m ³
Tensile Strength	290	N/m m ²
Compressive Strength	—	N/m m ²
Yield Strength	255	N/m m ²
Thermal Expansion Coefficient	2.38×10^{-5}	1/K
Thermal Conductivity	137	W/(m·K)
Specific Heat	880	J/(kg ·K)
Material Damping Ratio	—	—

Figure 4. Material Properties Table for 5052 H38 Alloy

Following the construction of the plate, the U-channel was created based off of a base model with the following measurements:

- Base Width: 3.13 in
- Side flanges: 0.63 in
- Channel length: 1.52 in
- Lip thickness: .15 in

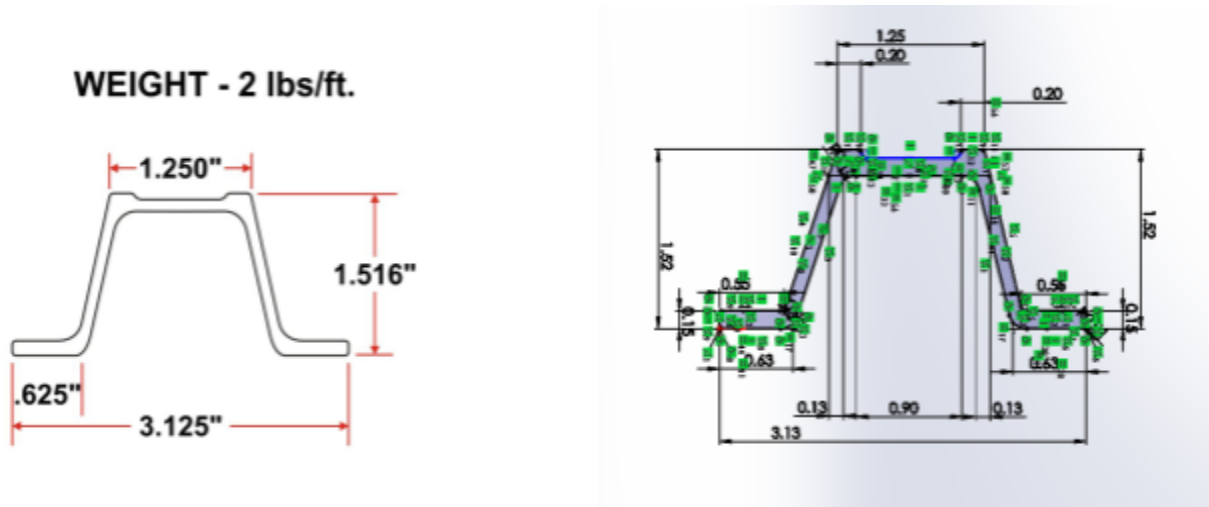


Figure 5. Reference U-Channel (left) SolidWorks U-Channel (right)

The post was then extruded to a height of 84 inches to replicate typical roadway stop sign installations. Followed by this, mounting holes were added which are normally used for attaching the actual plate or any regulatory sign. The mourning holes followed the following specifications:

- Hole diameter: .39 inches
- Hole spacing: 1 inch apart center point to center point.

Once the construction of the post was finalized, the material assigned to the post was galvanized steel.

Material Properties - Galvanized Steel

Property	Value	Units
Elastic Modulus	200,000	N/mm ²
Poisson's Ratio	0.29	—
Shear Modulus	—	N/mm ²
Mass Density	7,870	kg/m ³
Tensile Strength	356.9006745	N/mm ²
Compressive Strength	—	N/mm ²
Yield Strength	203.9432426	N/mm ²
Thermal Expansion Coefficient	—	1/K
Thermal Conductivity	—	W/(m·K)
Specific Heat	—	J/(kg·K)
Material Damping Ratio	—	—

Figure 6. Material Properties Table of Galvanized Steel

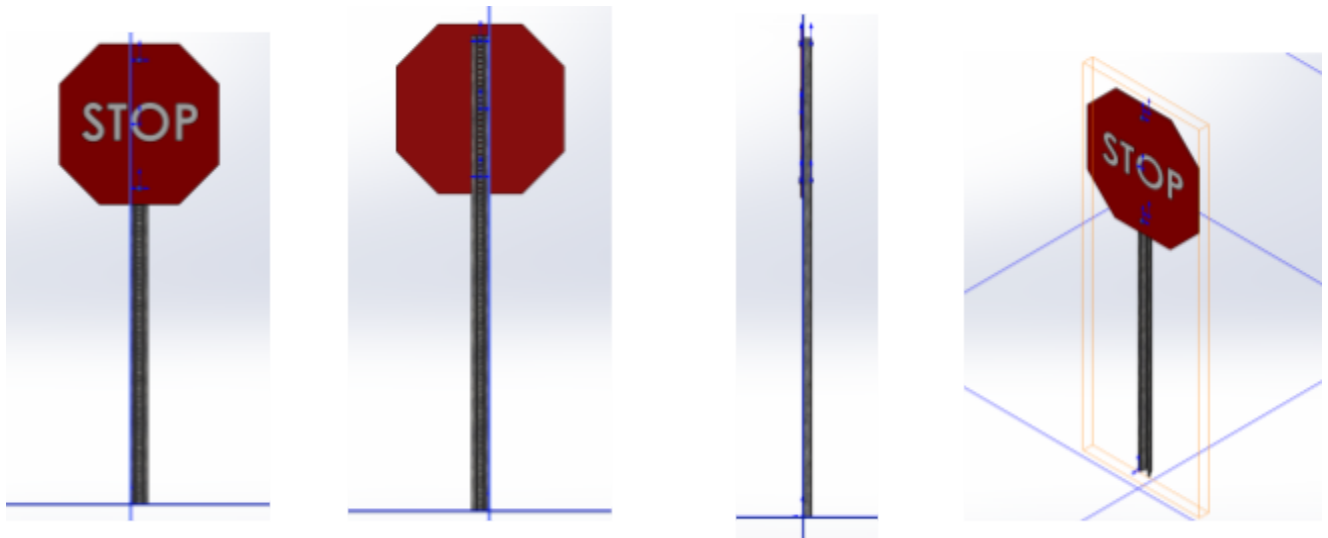


Figure 7. Full Stop Sign from different Angles

Simulations Methodology:

Fluid Dynamics Simulation (Finite Volume Analysis)

To analyze the stop signs response to behavior, an analysis was set up as an external time dependent flow problem, verifying it is still in the IPS unit system. The airflow around the stop-sign assembly was simulated in SolidWorks Flow Simulation using a project named **“Hurricane”**.

First a domain was created around the 30 x 30 inch sign and u-channel beam post with enough clearance in all directions so that the wake could form without interacting with boundaries. Air was selected as the working fluid and the simulation allowed for both laminar and turbulent flow. This is to account for transition behavior at higher wind speeds. The inlet boundary condition was defined as a uniform 110 mph hurricane category 2 strength wind which was converted to inches per second making it roughly 1939 in/s, directed perpendicular to the sign face. The air temperature was set to approximately 80°F to mimic typical hurricane conditions.

This simulation was run as a transient simulation over 30 seconds, A mesh was generated to capture with refinement near the signs edges, the wake region downstream, and the post to better resolve the pressure gradients and separated flow.

Over the 30 second simulation, performance metrics were defined within the Flow tree. The metrics, or goals, were defined to be the drag force which represents the force component parallel to the wind direction, the lift force Y which is the component normal to the flow and the lift to drag ratio. Non dimensional aerodynamic coefficients C_d and C_L . After the simulation run was completed, processing tools were used to visualize the pressure distribution on the sign. The final pressure on the sign surface from this simulation was then used as the applied load for the static simulations.

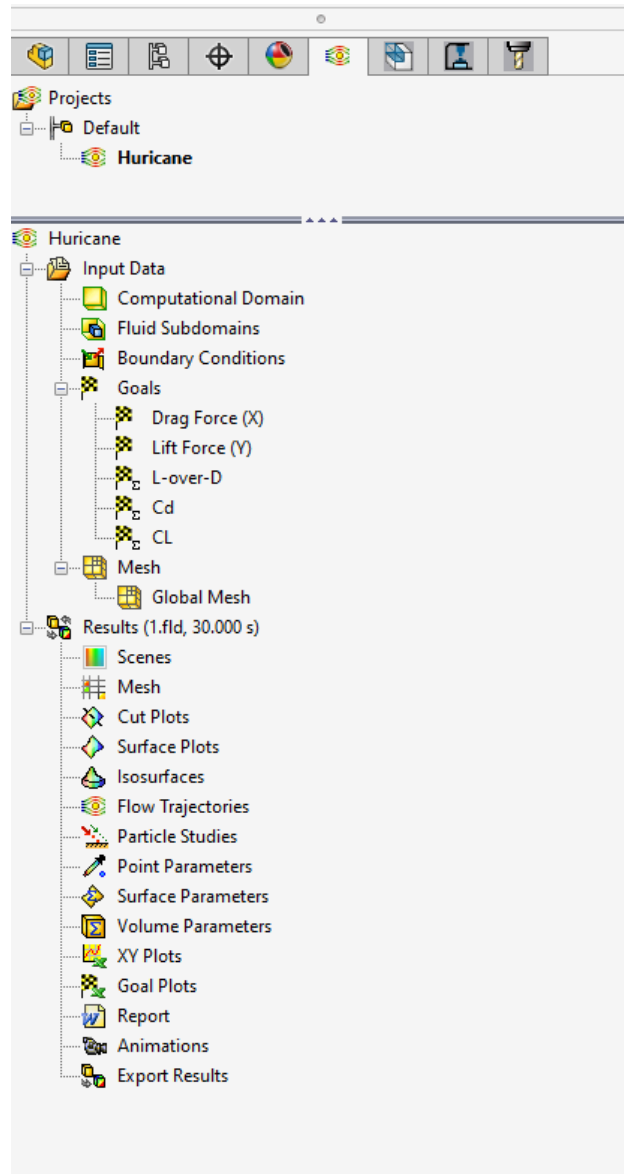


Figure 8. Flow Tree of Simulation 1.

Static FEA Analysis Static Structural Simulation Methods

In solid words simulation the static structural analysis of the stop sign assembly was run to evaluate the bending, deformation, stress and strain under terrain level wind loading. In order to mimic a realistic roadside installation of a stop sign the bottom of the U channel post was defined as a fixed support, restraining all translation and rotational degrees of freedom. This assumption is to model realistic conditions such as if the post were in concrete or soil. The aerodynamic drag calculated from the flow simulation was an avenger of 206.86 lbf which directly applied to the windward face of the stop sign plate.

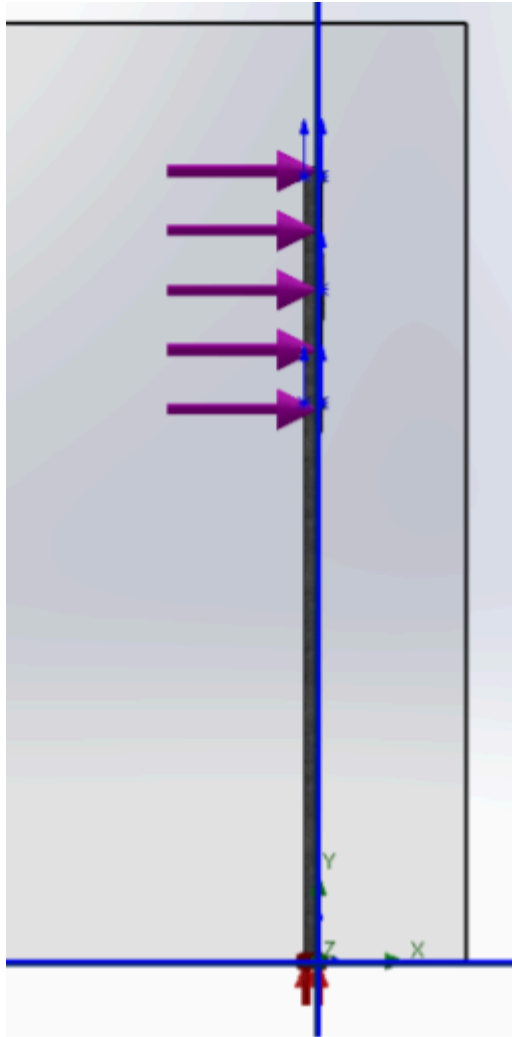


Figure 9. Direction of Force on Sign plate For Simulation 2

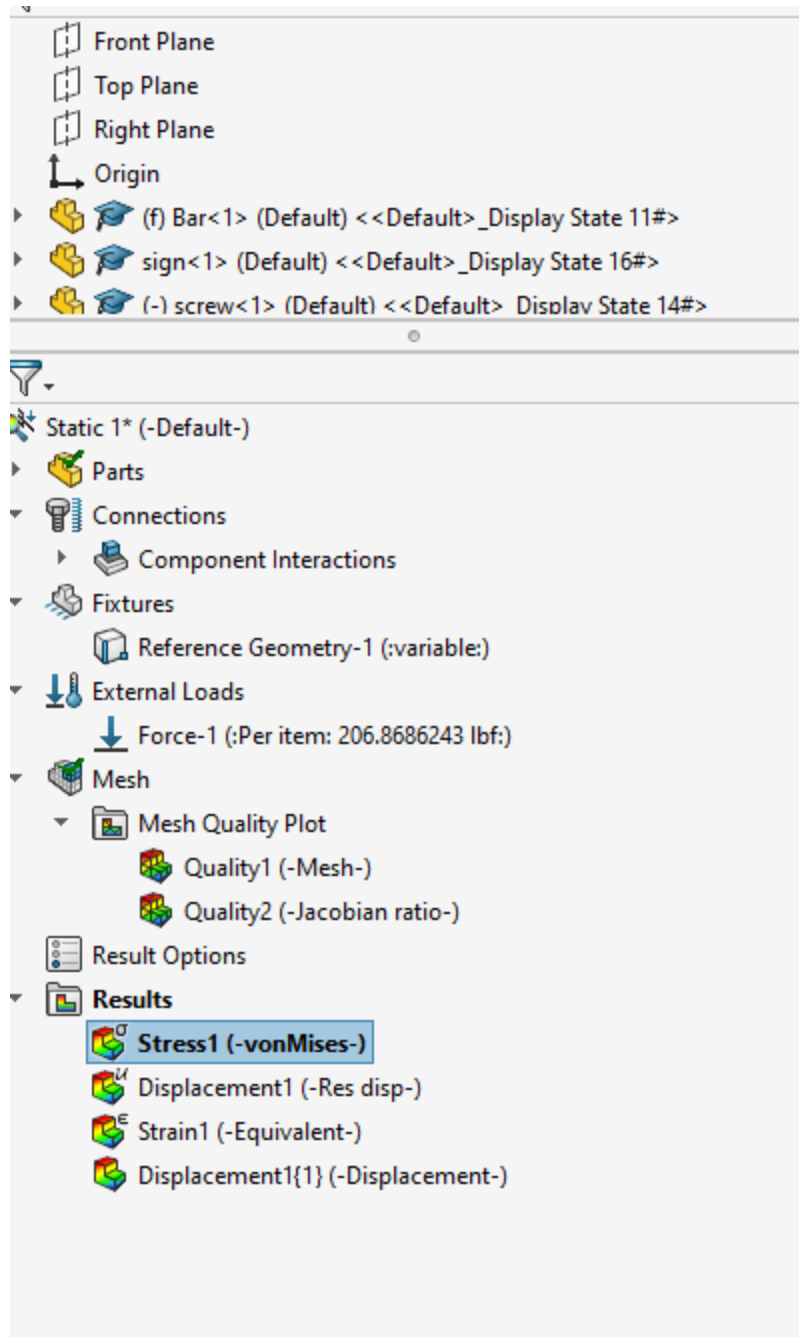


Figure 10. Flow Tree from Simulation 2

The stop sign assembly mesh was created using a curvature based solid mesh with refinements in places expected to stress concentration. This included the two mounting holes on the plate, the plate post interference and the bends and thinner regions of the U-channel cross section.

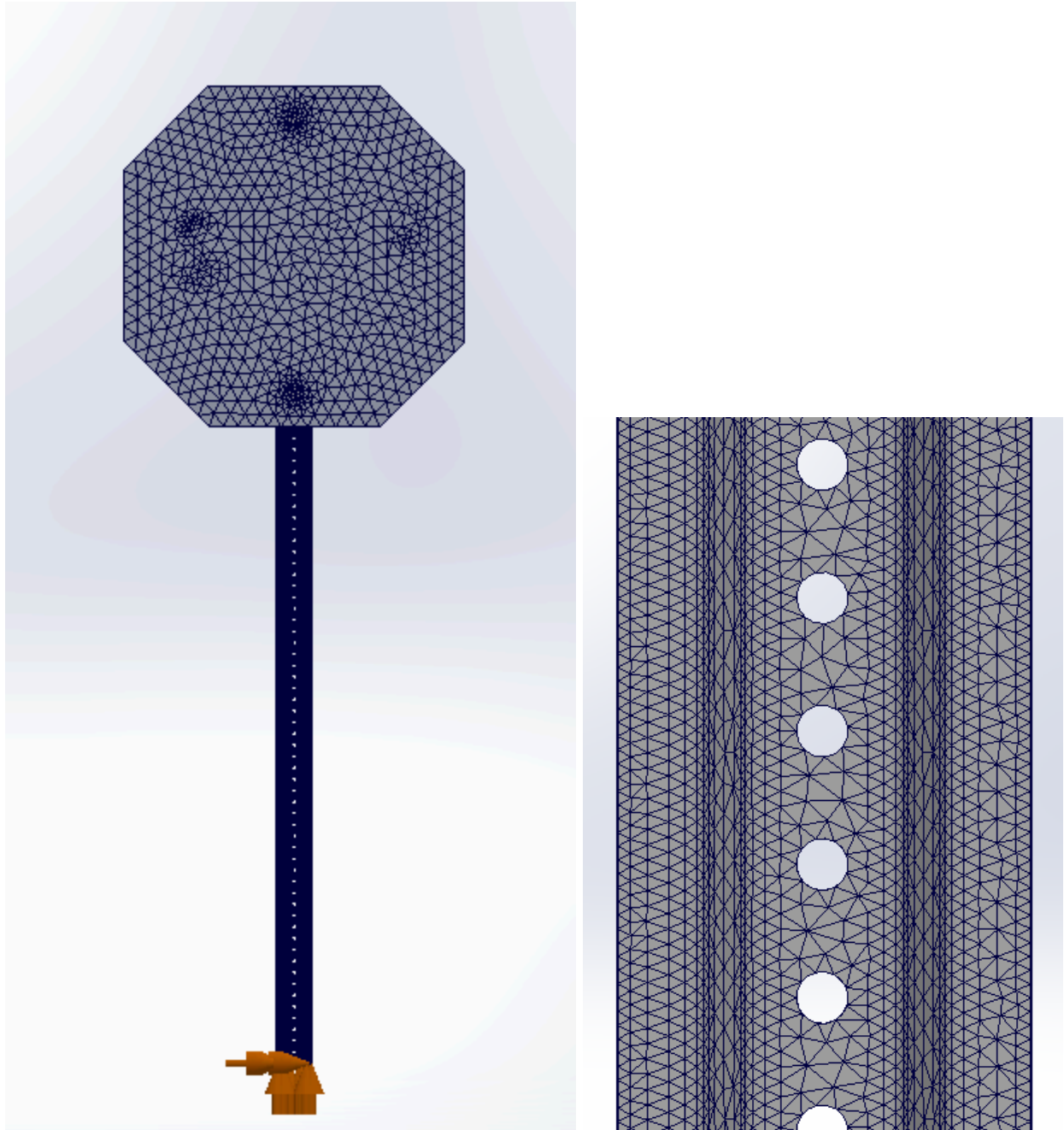


Figure 11. Stop Sign Mesh for Simulation 2

This allows for accurate capture of the bending stresses and localized strain behavior. The static solver was run under linear elastic assumptions in order to evaluate the stop signs response to the applied wind load. This analysis computed the displacement field, the bending of the steel channel post and the aluminum sign plate. After the simulation was completed the following output results were generated:

- Von Mises Stress Plot
- Displacement Plot
- Strain contour

These results provided the structural response needed to evaluate the stop sign's performance under hurricane level aerodynamic loading.

FEA - Modal Analysis (Vibrations)

The last simulation run of the stop sign model was to determine the natural frequencies and vibration mode of the full assembly. The frequency study used the same curvature based tetrahedral mesh that was generated for the static analysis. In order to capture local deformation patterns and global bending modes of the post, refinement around the bolt holes, paste to post interface and perforation along the U channel were included in the mesh.

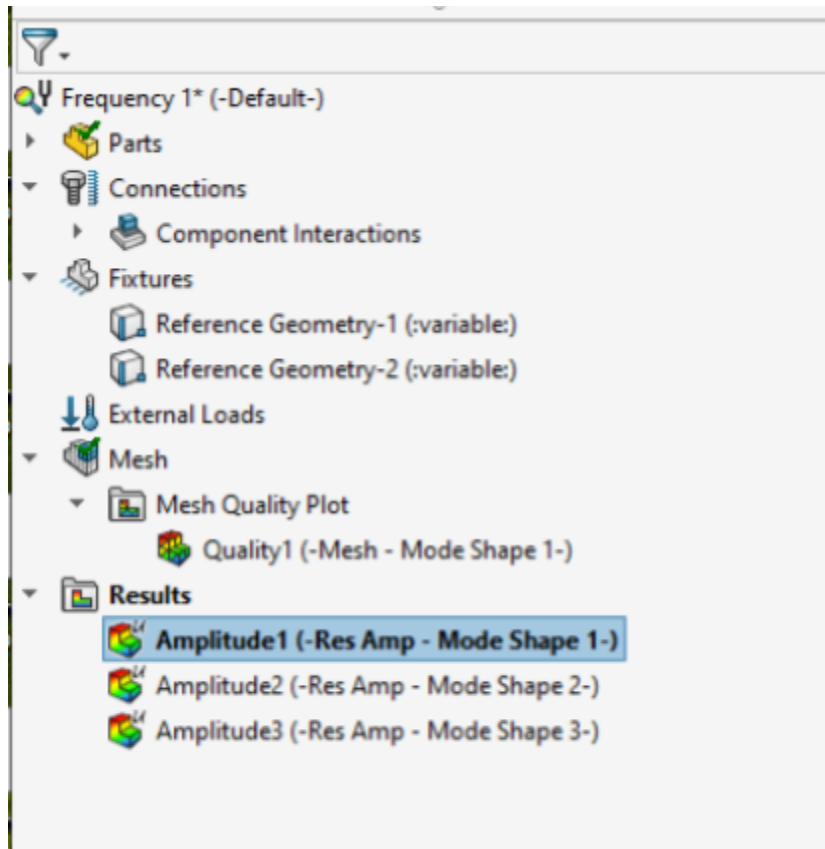


Figure 12. Flow Tree for Simulation 3

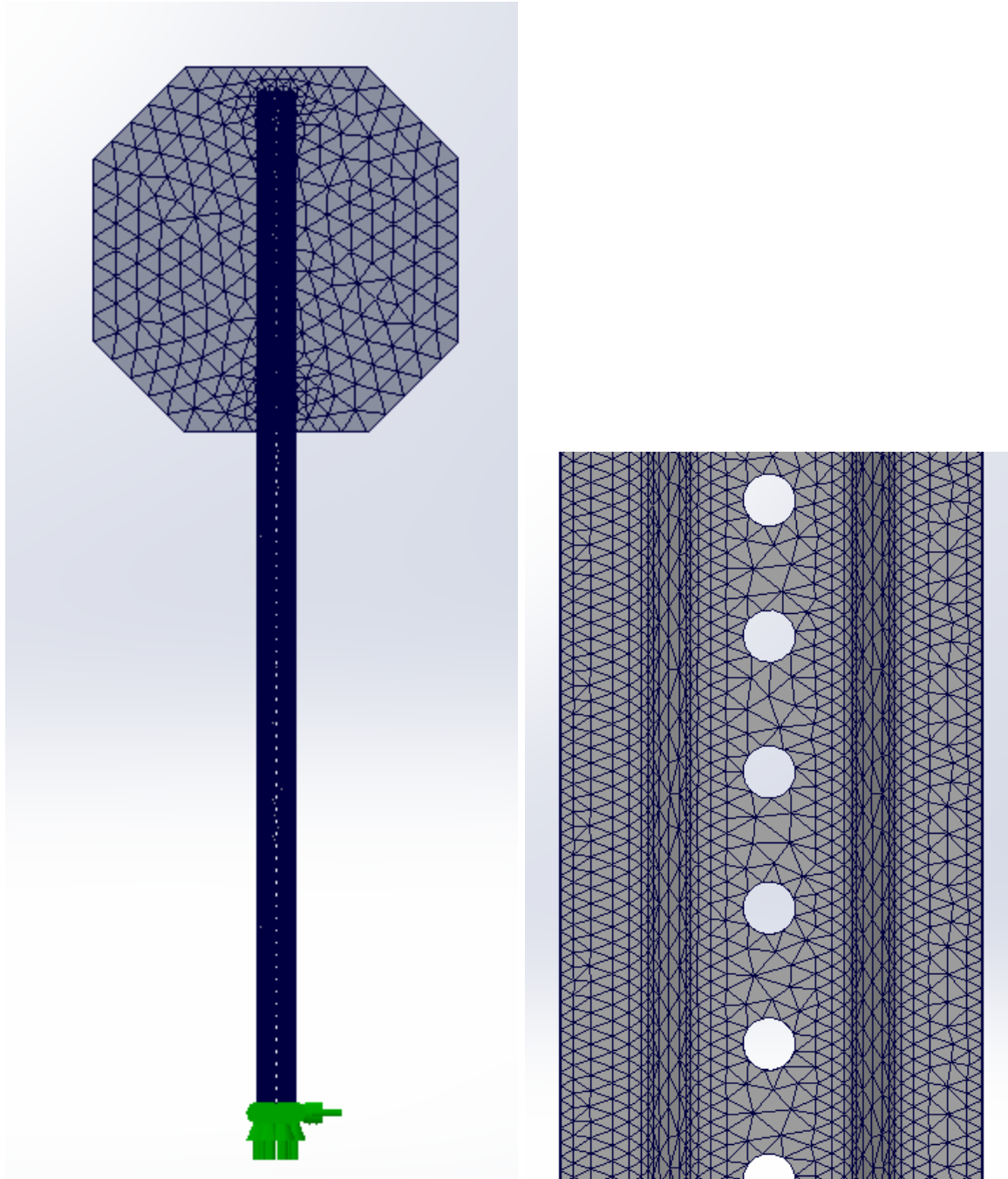


Figure 13. Mesh of Stop sign for Simulation 3

The bottom of the sign post was defined as a fixed support similar to the static analysis. There were no external loads applied during this study as it solves the free vibration eigenvalue problem configured to extract the first three natural requests and corresponding mode shapes.

After the simulation was run, the output reported the frequency in both rad/s and Hertz. In addition each mode was visualized using the deformation scaled contour plots showing modal

amplitude across the plate and post. The regions which undergo the greatest relative motion in each vibrational mode are highlighted in the contour plots.

V. Results and Discussion

Results:

Fluid Dynamics Simulations:

The aerodynamic and structural response of the stop sign under hurricane of category 2 level wind loading was evaluated by running a simulation with defined performance metrics. The following table summarises the converged values for drag force, lift force, L/D ration, Cd and CL.

Goal Name	Unit	Value	Averaged Value	Min. Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria	Maximum in Calculation	Time of Maximum [s]
Drag Force (X)	[lbf]	206.8686243	206.7597112	206.6863341	206.8686243	100	Yes	0.001924603	74.01546348	2526.404904	0.000147132
Lift Force (Y)	[lbf]	-0.07110569	-0.070726072	-0.072388353	-0.06932977	100	Yes	0.000274218	0.044191692	0.516772432	0.001584609
L-over-D	[lbf]	-0.00034372	-0.000342069	-0.000350079	-0.000335328	100	Yes	1.3298E-06	9.01727E-05	0.000962745	0.001584609
Cd	[-]	0.000935106	0.000934614	0.000934282	0.000935106	100	Yes	8.69976E-09	0.000334571	0.011420079	0.000147132
CL	[lbf]	-3.2142E-07	-3.19702E-07	-3.27216E-07	-3.13391E-07	100	Yes	1.23954E-09	1.99759E-07	2.33596E-06	0.001584609

Figure 14. Summary Table of Output Result of Simulation 1

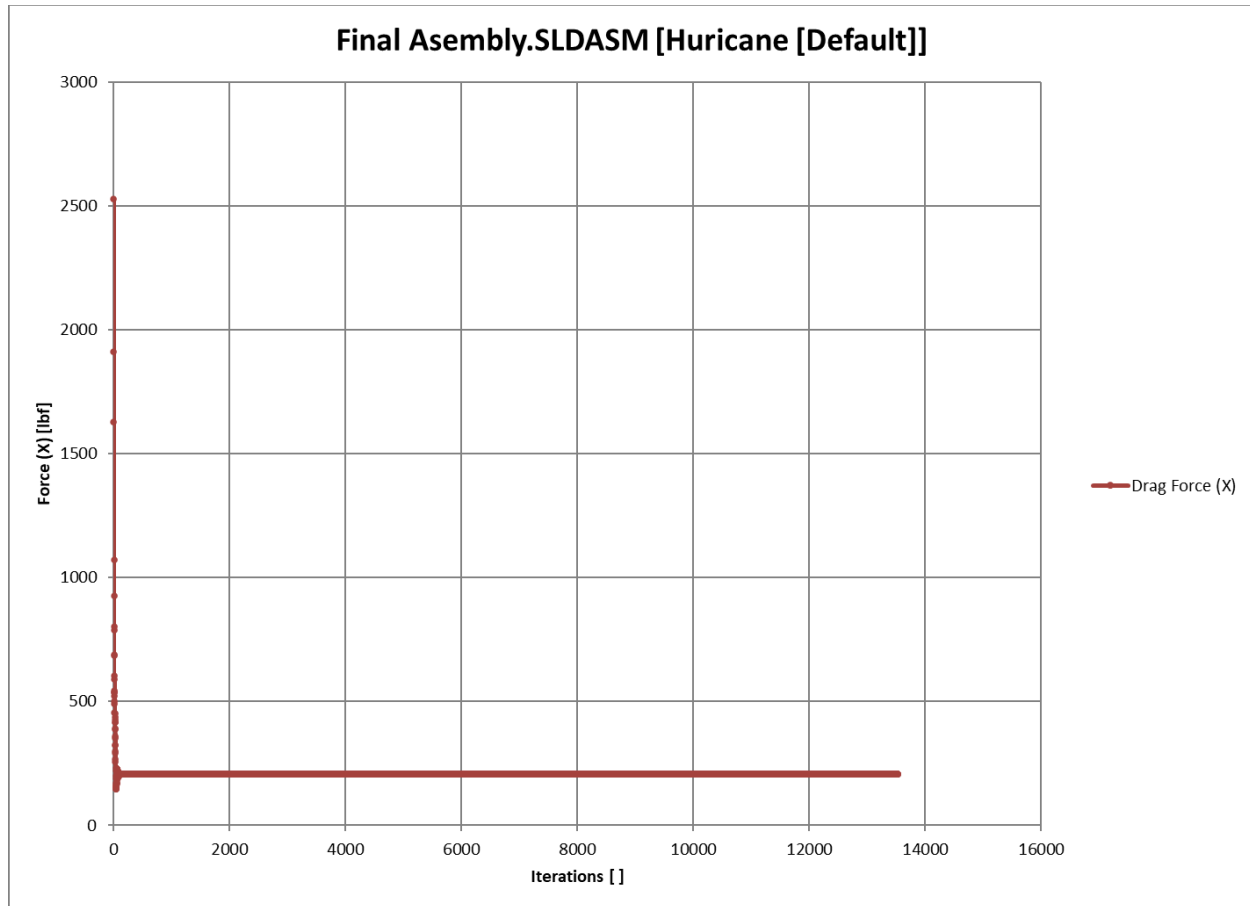


Figure 15. Drag force Plot from Simulation 1

The sign experienced a steady drag force of about 206.86. The drag force goal plate showed a brief surge during the first 200 iterations followed by a rapid settling to around 206 lbf for the remaining simulation.

The lift force essentially remained zero, it has a converged magnitude of approximately -0.07 lbf. The lift plot demonstrates only small numerical oscillation in the order of 10^{-2} to 10^{-3} lbf before settling barely on the zero line. This is expected for a flat plate oriented normal to the flow, where the pressure distribution is symmetric about the vertical axis. This produces no meaningful upward or downward force.

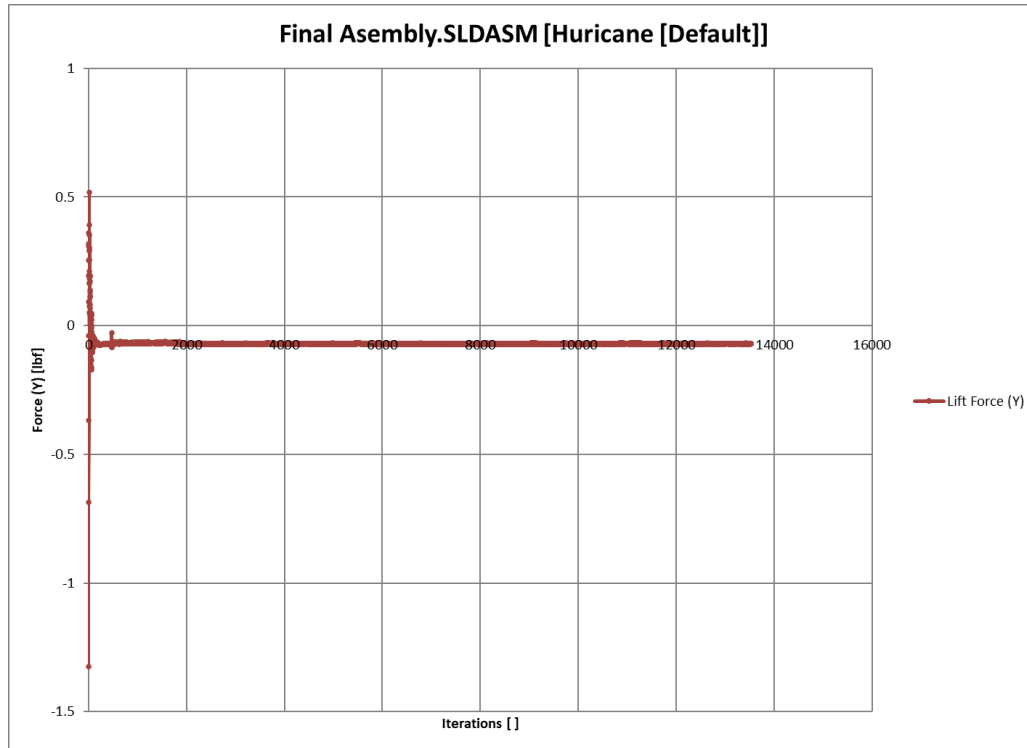


Figure 16. Lift Force Plot from Simulation 1

The drag coefficient converged to 0.000935 and the lift coefficient converged to approximately -3×10^{-7} . Both showed clean convergence behavior with values stabilizing after the initial flow development period. The small CL confirms the absence of vertical aerodynamic loading, while the Cd value is low due to the unit system and scaling of the area it remains constant and steady with the reported drag forces

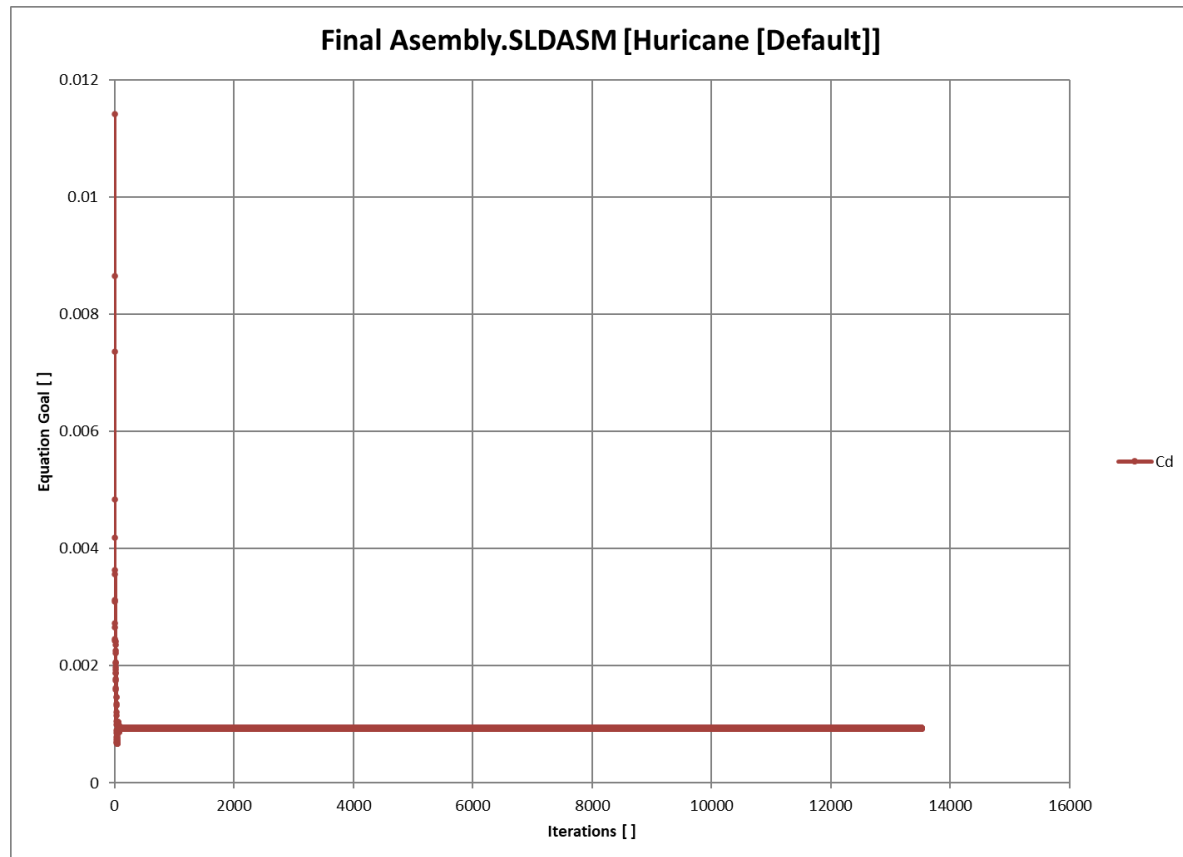


Figure 17. Drag Coefficient Plot from Simulation 1

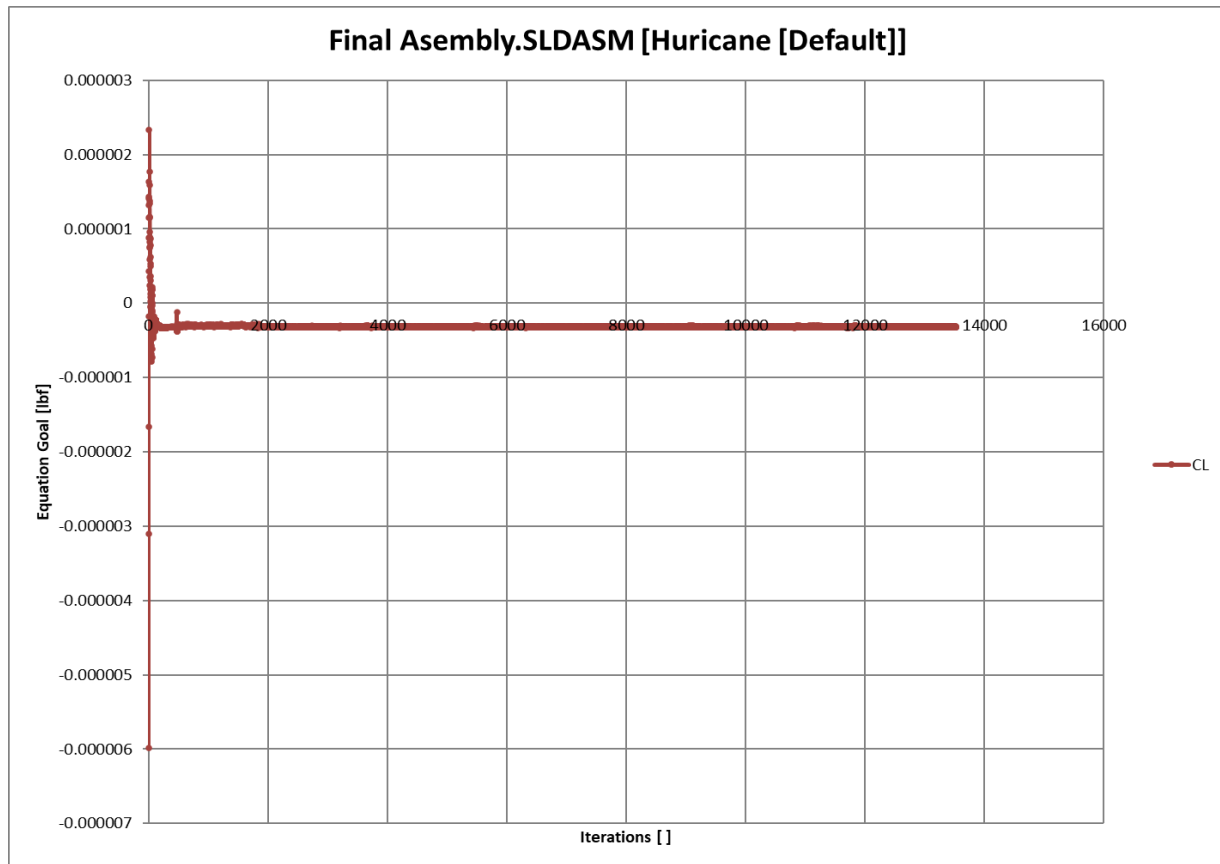


Figure 18. Lift Coefficient Plot from Simulation 1

The lift to drag ratio remained near zero with a converged value of -0.000343. This reflects negligible life relative to the dominant drag.

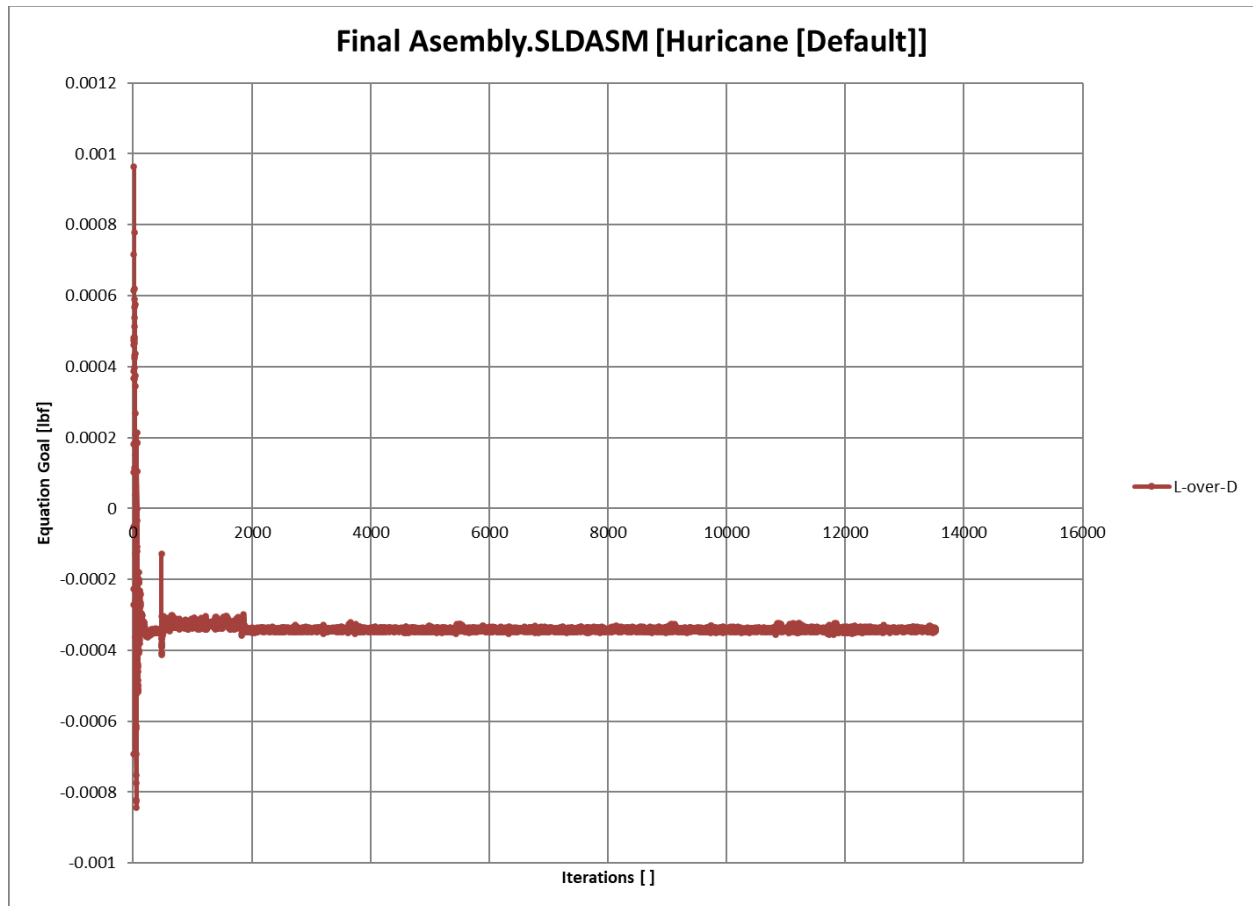


Figure 19. L-over-D ratio Plot from Simulation 1

The flow trajectory visualizations give further insight into the aerodynamic behavior of the system itself. The side view pressure map shows a high pressure stagnation region on the windward face of the sign.. Flow separates cleanly around the edges of the plan which forms a board wake downstream. The post lies in this region so it experiences reduced dynamic pressure and lower velocity compared to the free stream.

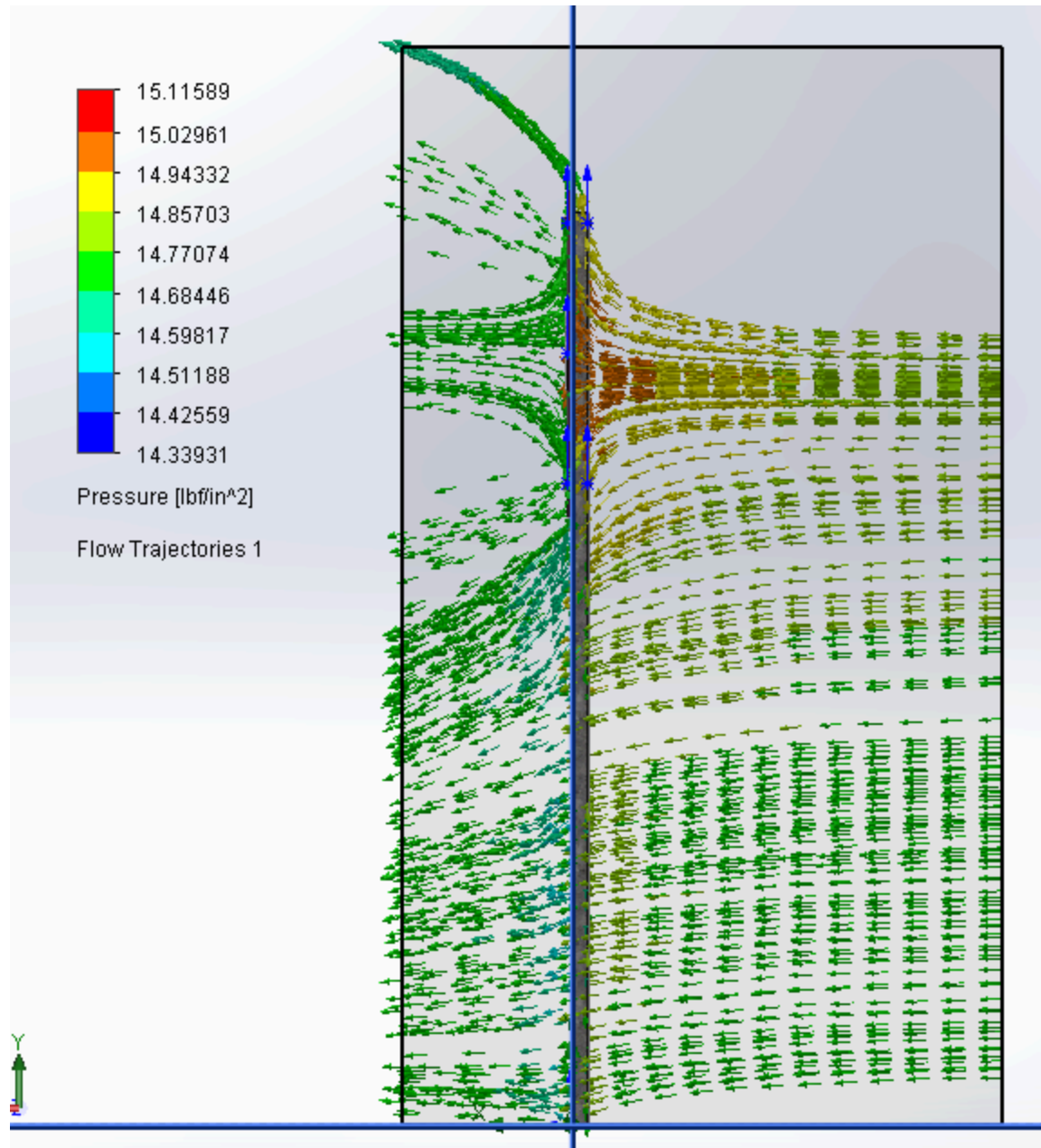


Figure 20. Sideview Visualization from Simulation 1.

The front view visualization showcases axisymmetric radial spreading of streamlines around the sign's feet, confirming asymmetric separation and the absence of any lateral or vertical bias. This aligns with the negligible lift observed in the goal plots.

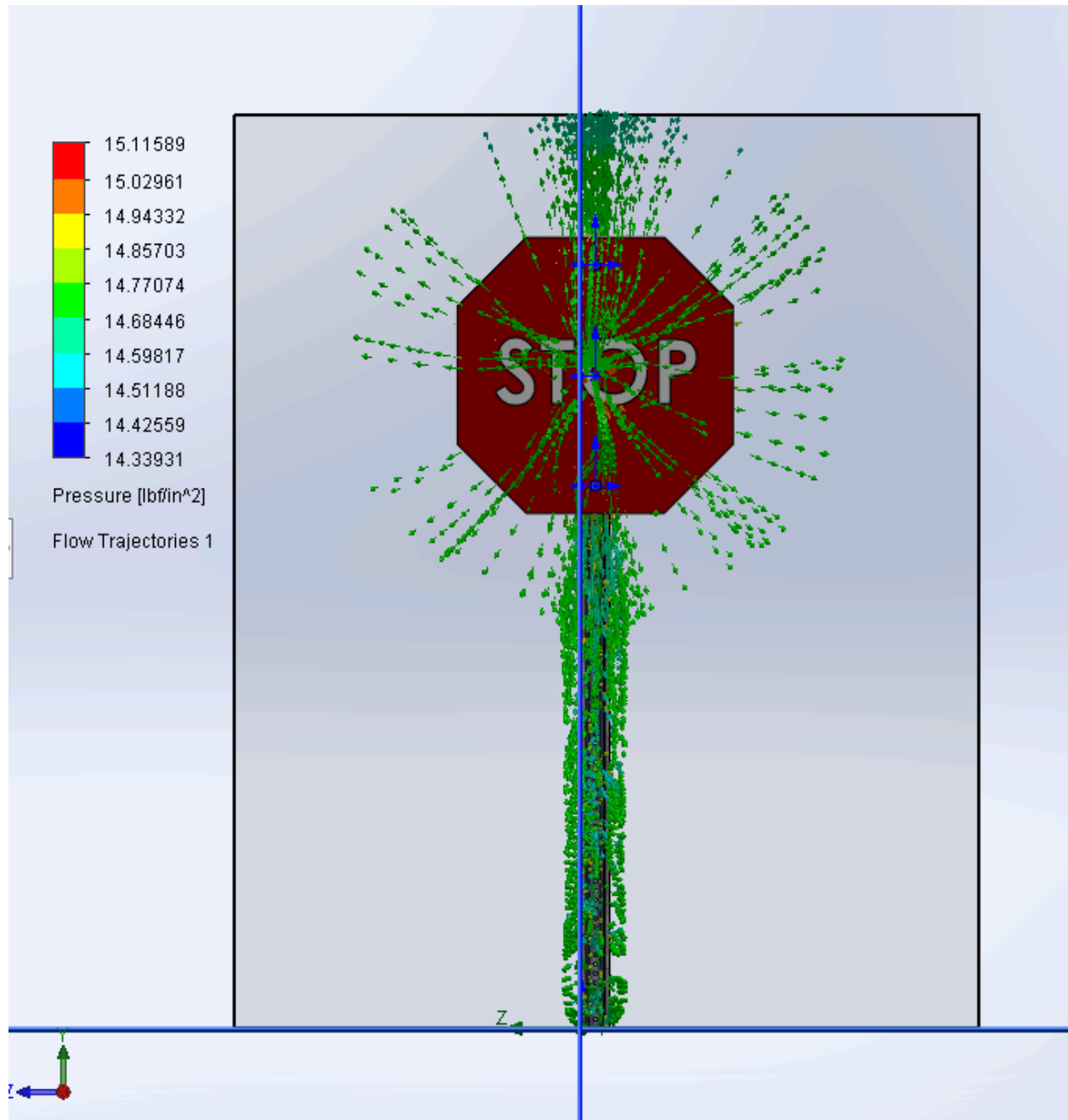


Figure 21. Frontview Visualization from Simulation 1

The angle view below demonstrated the full wake structure. The flow accelerates around the plate edges producing a characteristic shear layer roll up and an extended recirculation zone behind the sign. This lower pressure wake area is responsible for the steady drag value

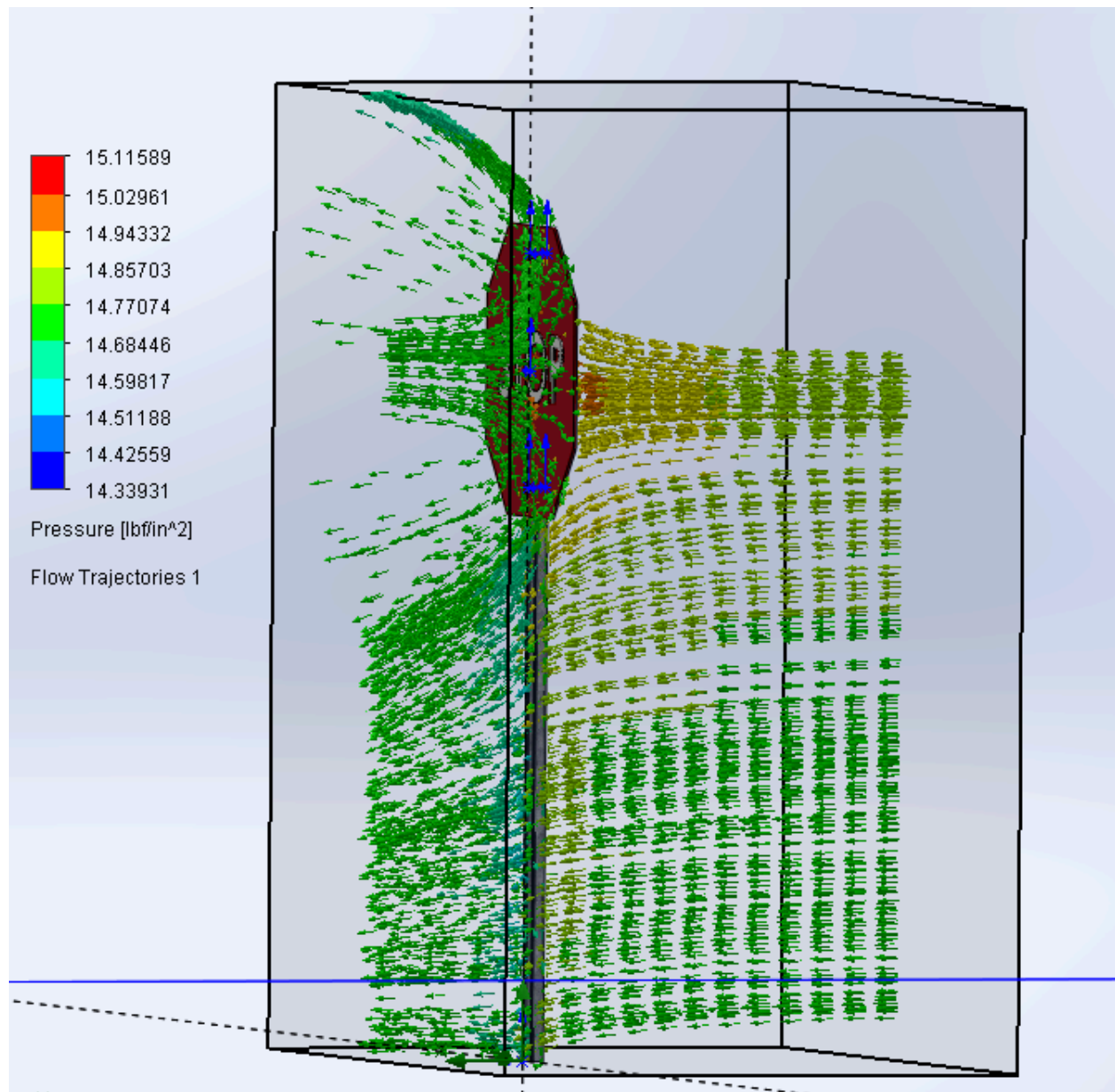


Figure 22. Angled visualization from Simulation 1

FEA Static

The static fine element analysis evaluated the structural response of the stop sign under the hurricane-induced drag of 206.86 lbf. This number was obtained from the fluid dynamic analysis performed previously. The force was applied to the sign face while the bottom of the U channel post was contained to present an in ground installation. Below is a summary of the minimum and maxim result from the Static FEA .

Quantity	Minimum	Maximum	Notes
Von Mises Stress (psi)	3.25 psi	123,900 psi	Min on sign face, max at base of post
Displacement (mm)	~0	145.3 mm	Max at sign's top edge
Strain (unitless)	1.608×10^{-7}	3.032×10^{-3}	Max at post base

Figure 23. Table Summary of Minimum and Maximum values from Simulation 2

The von Mises stress contours show that maximum stress occurs at the fixed base of the post with a peak value of 123,900 psi. The lowest stress appears along the face of the sign with a minimum of 3.25 psi indicating that the sign plate carries very little structural layout relative to the post. The contours outputted the min and max in Newtons but were converted to psi.

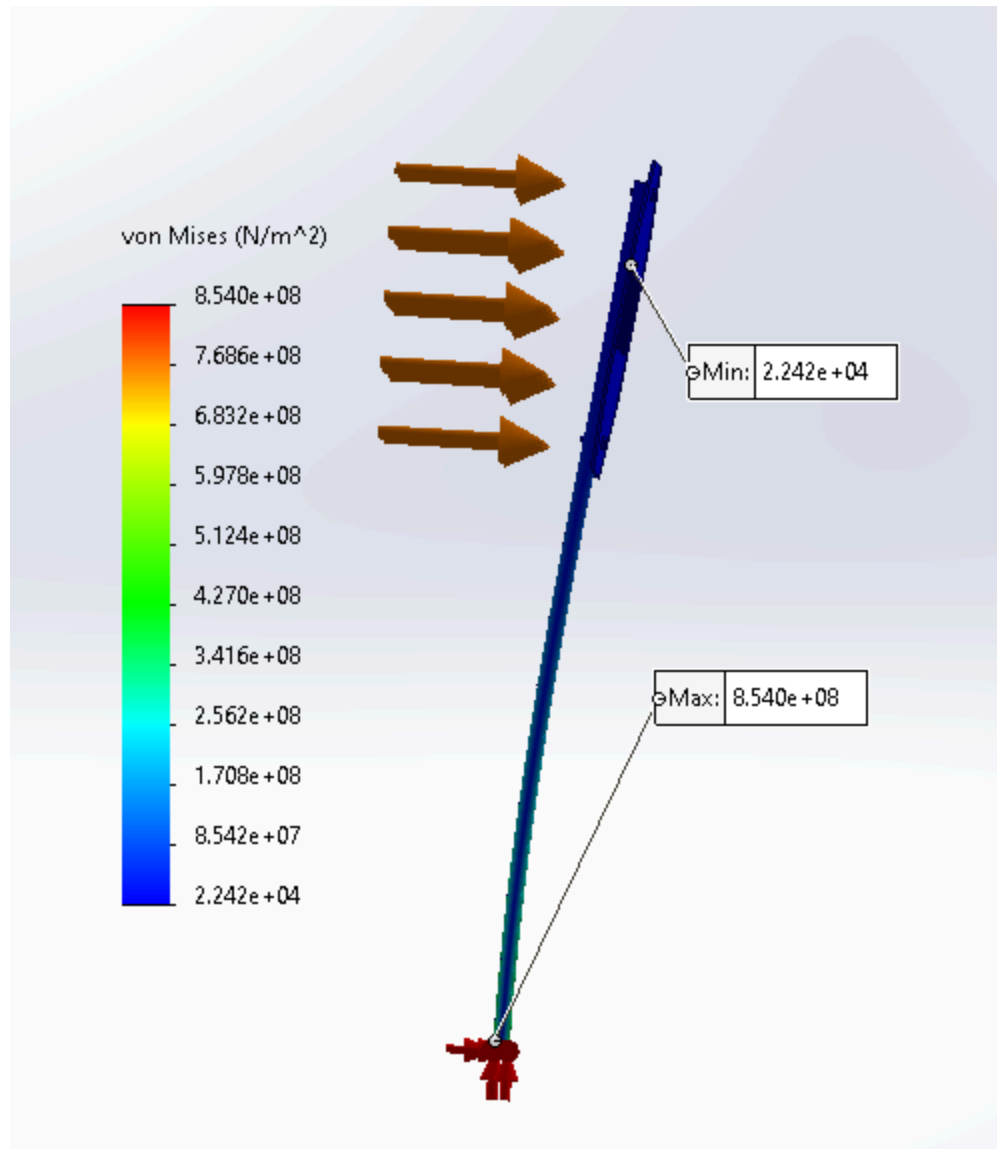


Figure 24. Von Mises Stress Contour Sideview

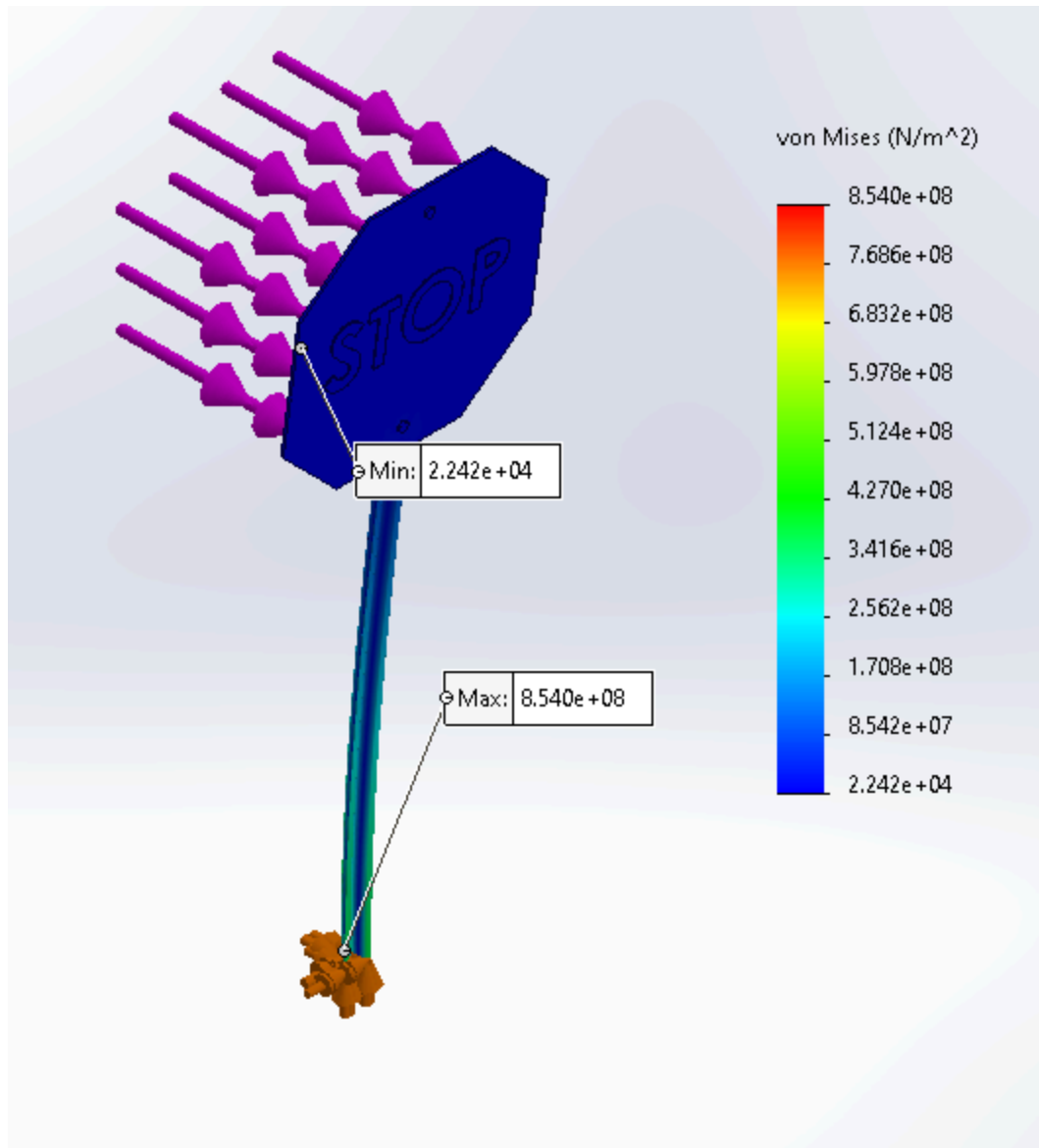


Figure 25. Von Mises Stress Contour Angled View

The stop sign structure demonstrated substantial bending under the hurricane level force. The maximum displacement of around 5.7 inches (converted from 145.3 mm) occurred at the top edge of the stop sign. The displacement at the fixed base is effectively zero which is constant with the boundary condition of it being fixed. The stop sign smoothly increases deflection as it moves up the post.

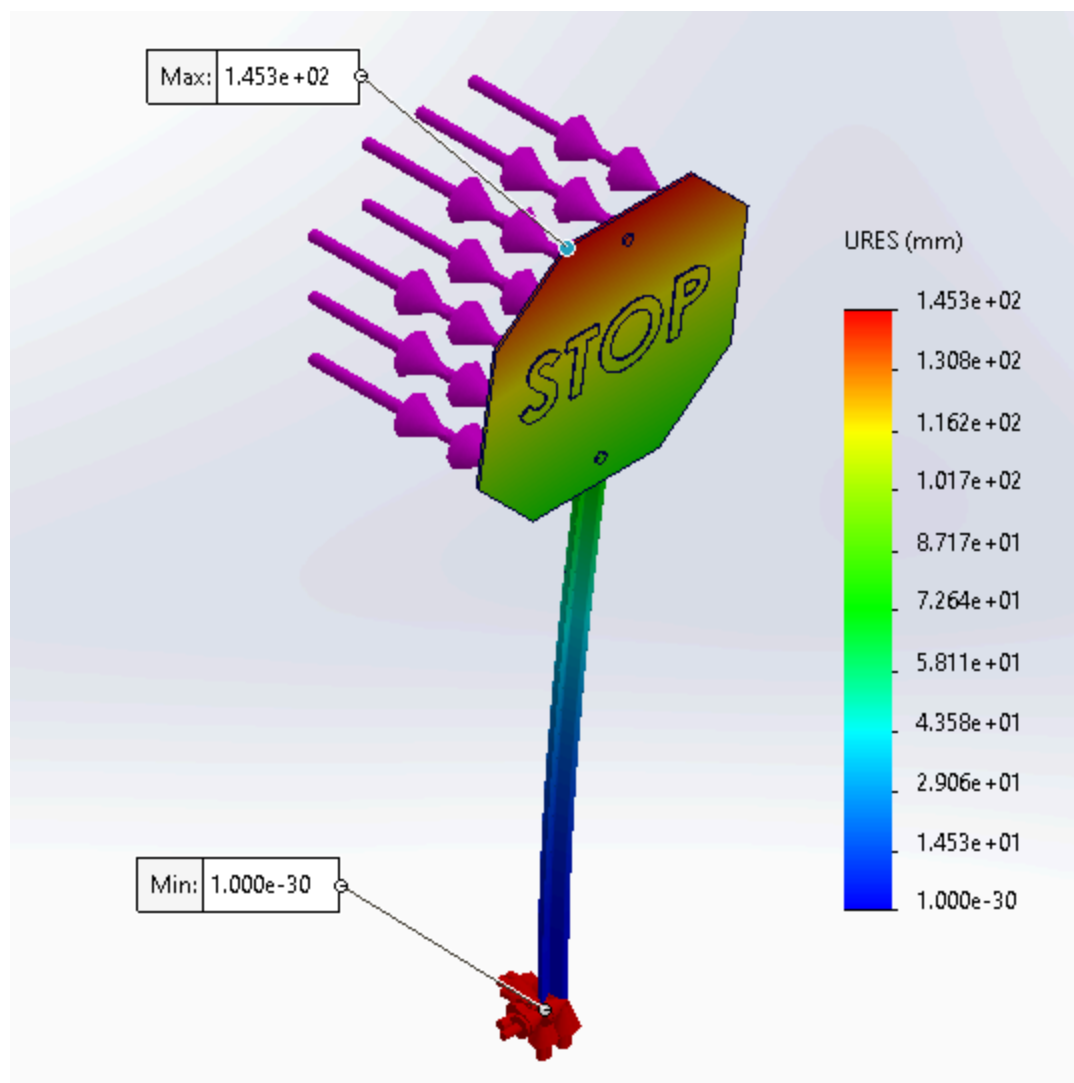


Figure 26. Displacement Contour Angled View

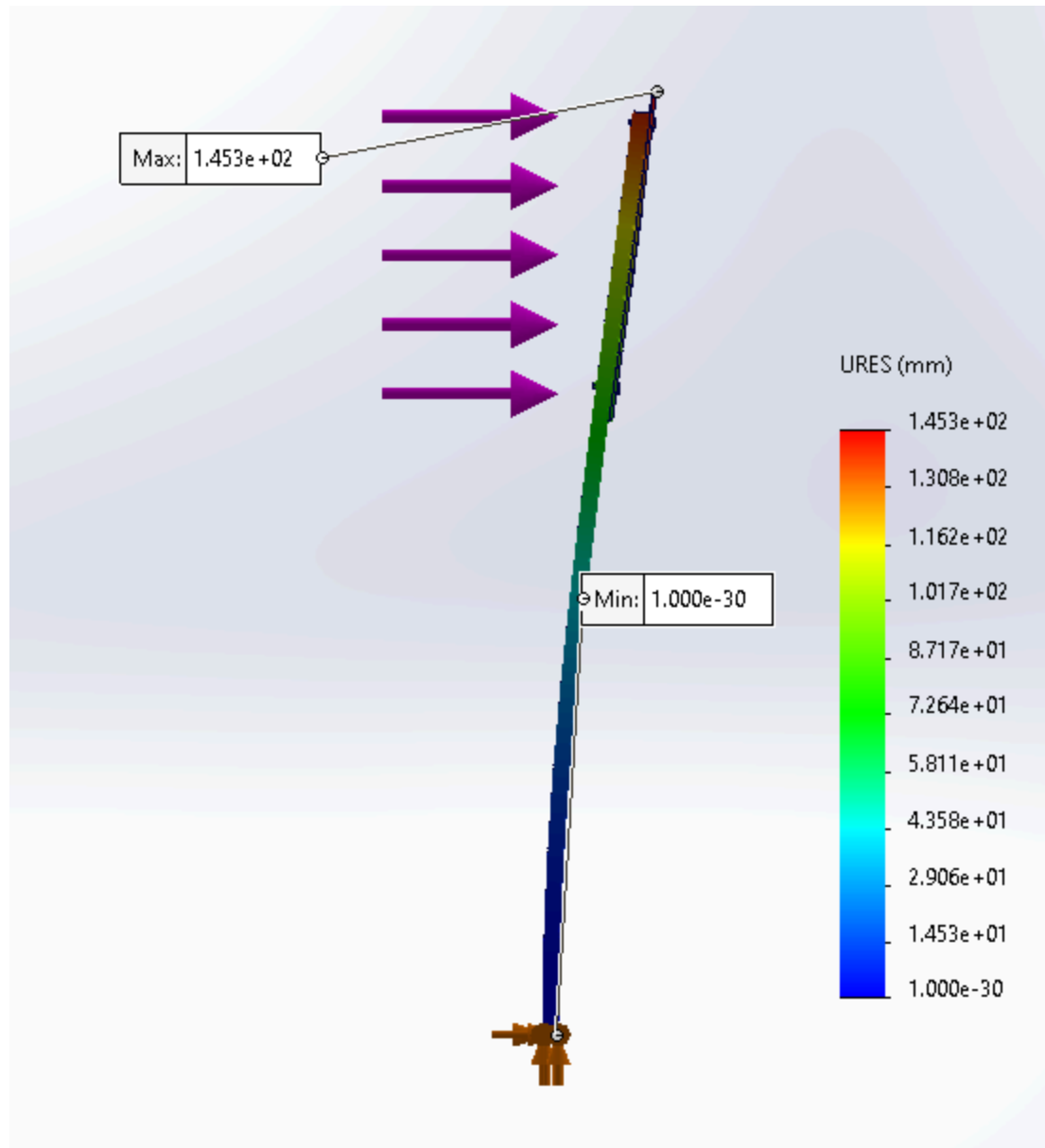


Figure 27. Displacement Contour Side view

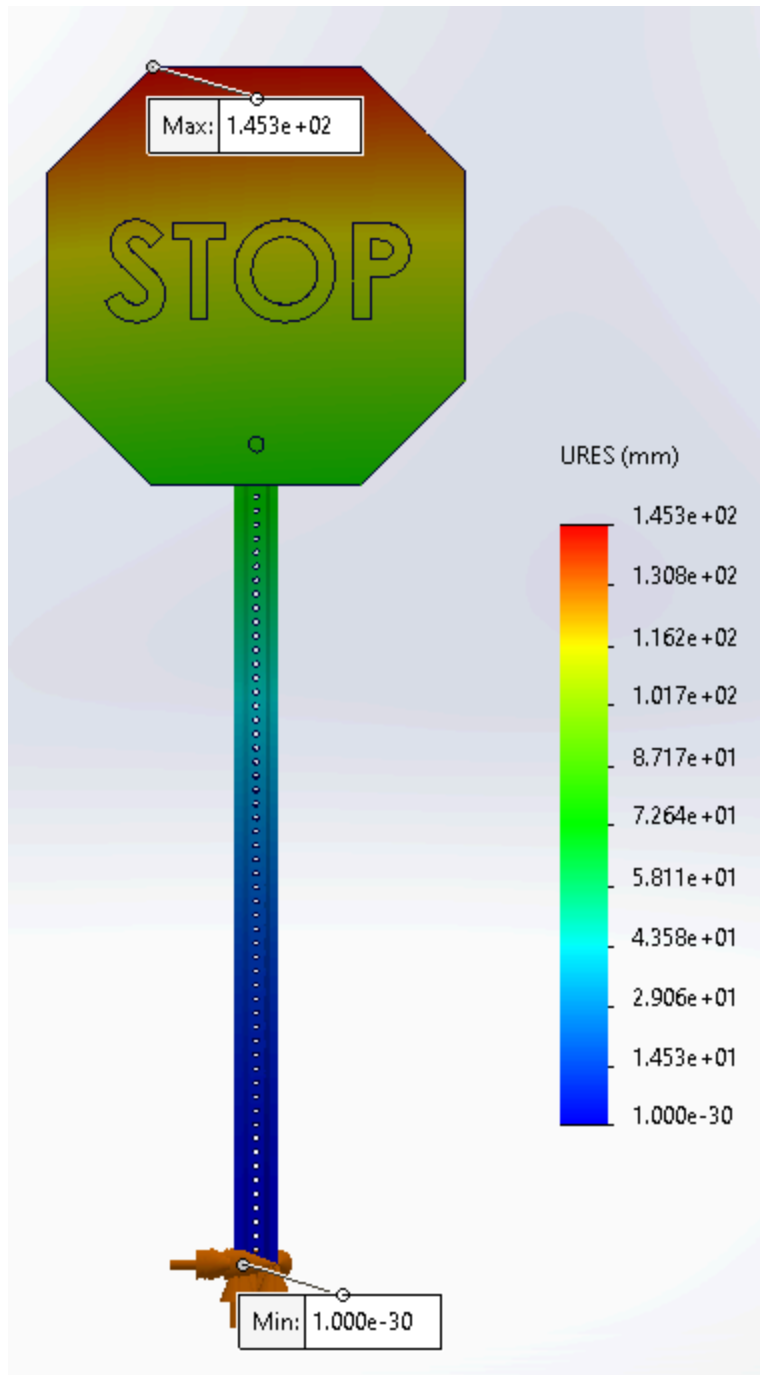


Figure 28. Displacement Contour Front View

The strain contour closely mirrors the prior stress field. The maximum strain was located near the base at 3.03×10^{-3} tapering upward along the pole. The minimum strain was 11608×10^{-7} on the face of the sign.

The majority of the structural demand is actually concentrated on the steel post as confirmed by the strain output of the simulation , while the sign plate only undergoes minor elastic deformation.

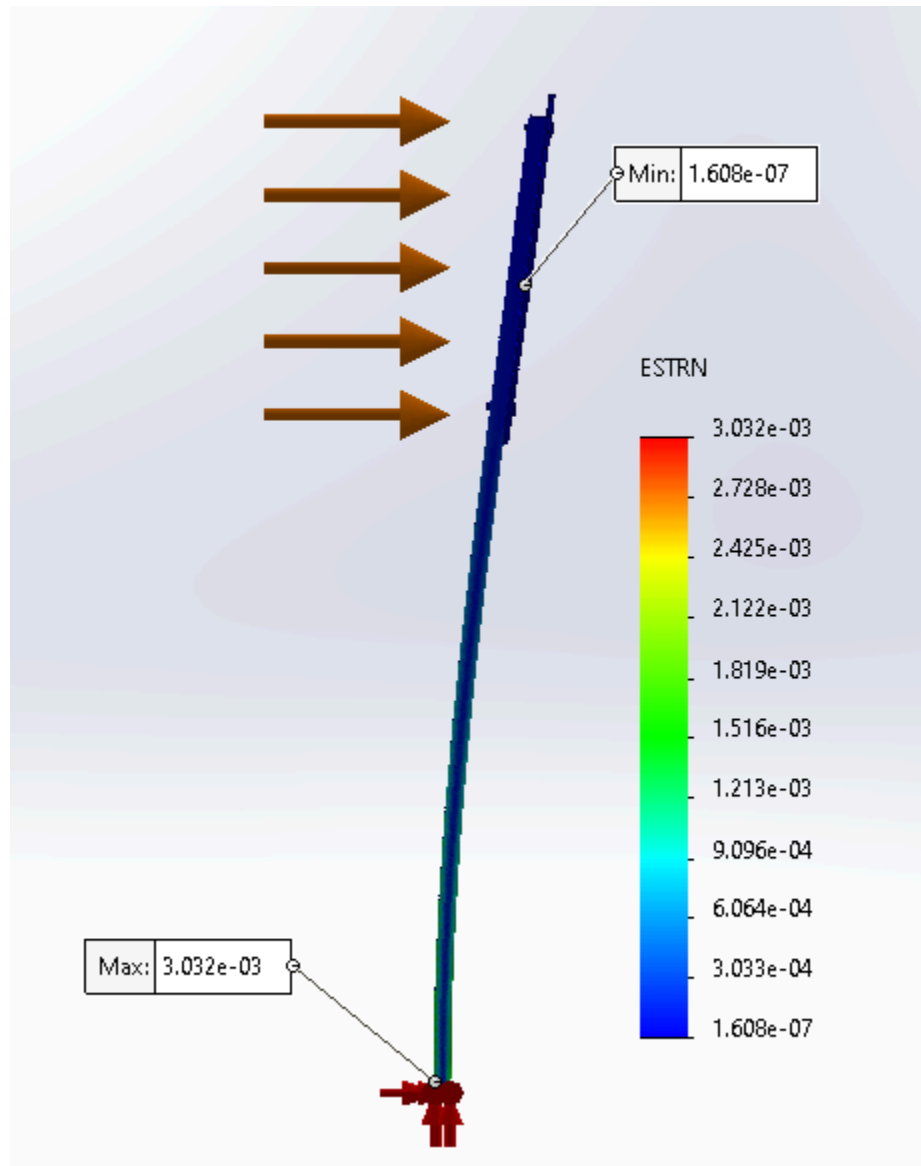


Figure 29. Strain Contour Sideview

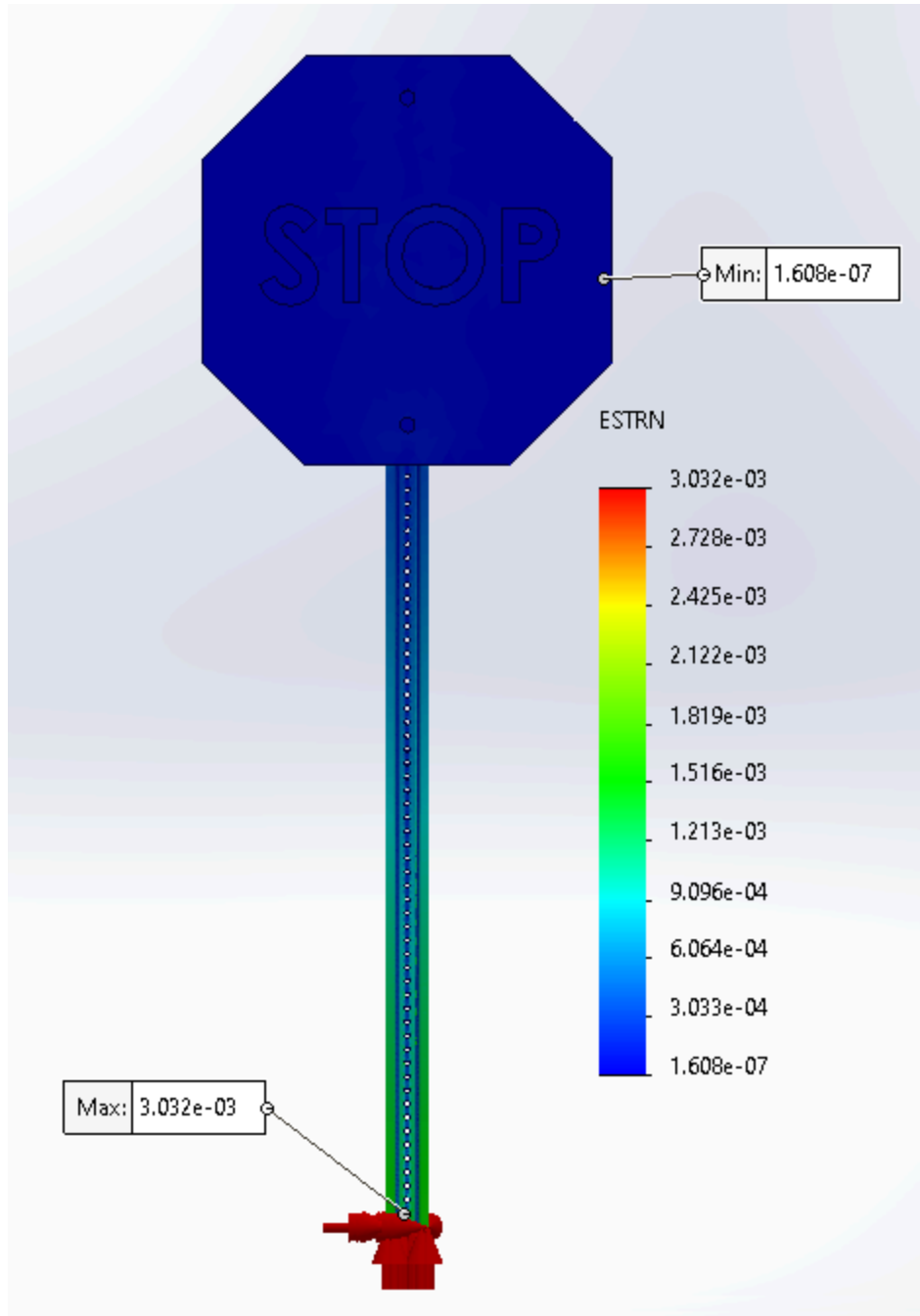


Figure 30. Strain Contour Front View

Vibrations FEA

A modal analysis was conducted to determine the natural frequencies and corresponding mode shapes of the stop sign assembly. Because modal analysis solves the free vibration eigenvalue problem, no external loads were applied; the solution depends solely on the geometry, mass distribution, and stiffness of the aluminum sign plate and galvanized steel U channel post.

The base of the post was fully fixed to represent in-ground installation. The mesh used in the frequency study matched the refinement strategy of the static analysis, ensuring accurate capture of bending modes while maintaining good element quality.

The solver extracted the first three natural frequencies. These low order modes are the most relevant for assessing susceptibility to wind-induced oscillations or vortex shedding.

Mode Shape 1: 0.5935 Hz (Fundamental Bending Mode)

Mode Shape 1 represents the dominant global bending of the structure. The maximum amplitude occurs at the upper edge of the sign plate, while the fixed base shows zero displacement, as expected. The entire post bends in a single smooth curvature pattern, characteristic of the first bending mode of a tall, slender cantilever.

The sign plate moves laterally in phase with the top of the post, acting as a rigid mass at the tip. This fundamental bending frequency of 0.5935 Hz is low due to the tall, flexible steel post and relatively light sign. Such low-frequency modes are most likely coupled with quasisteady wind loading and large-scale wake oscillations.

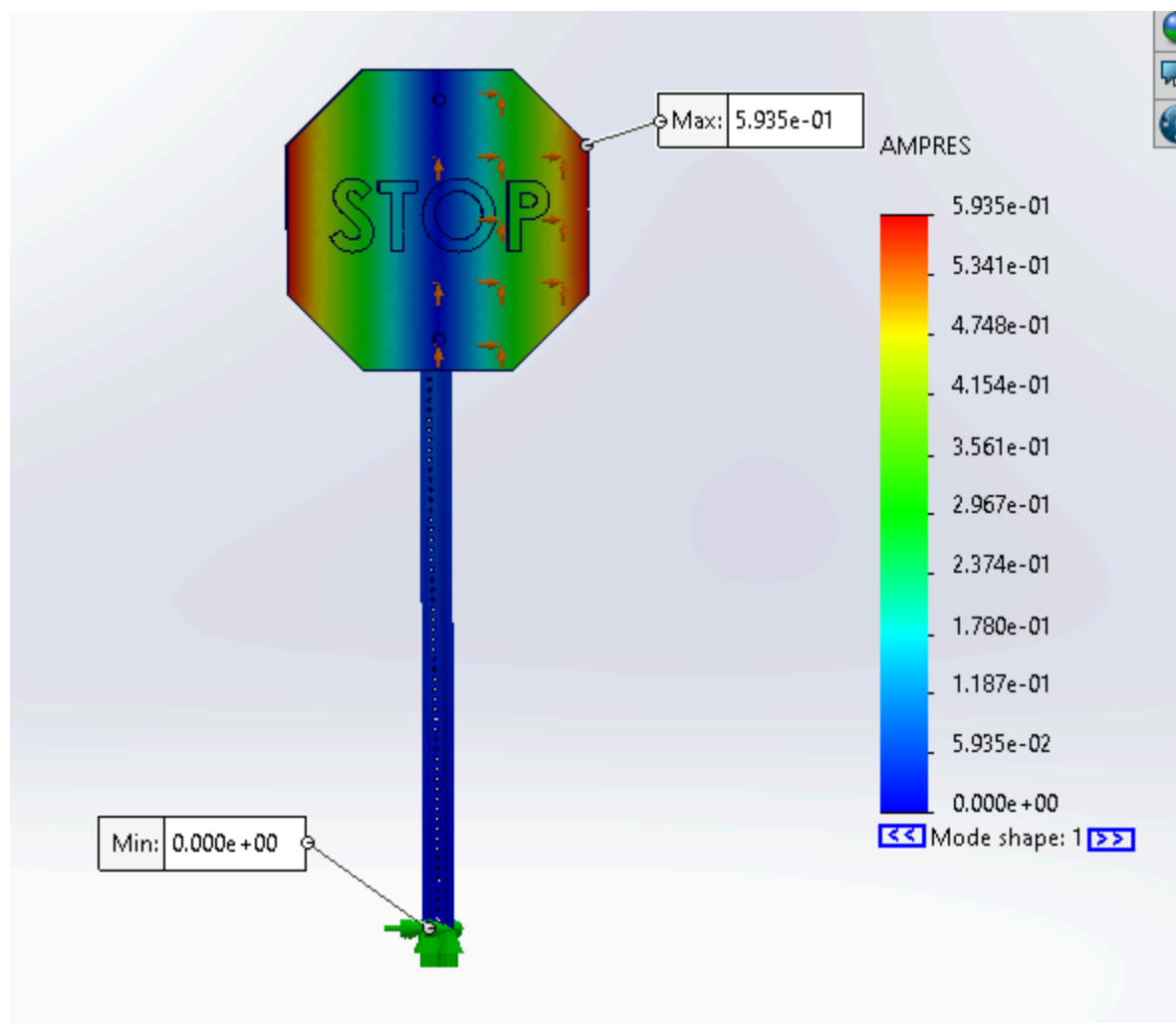


Figure 31. Mode 1 Front view from Simulation 3

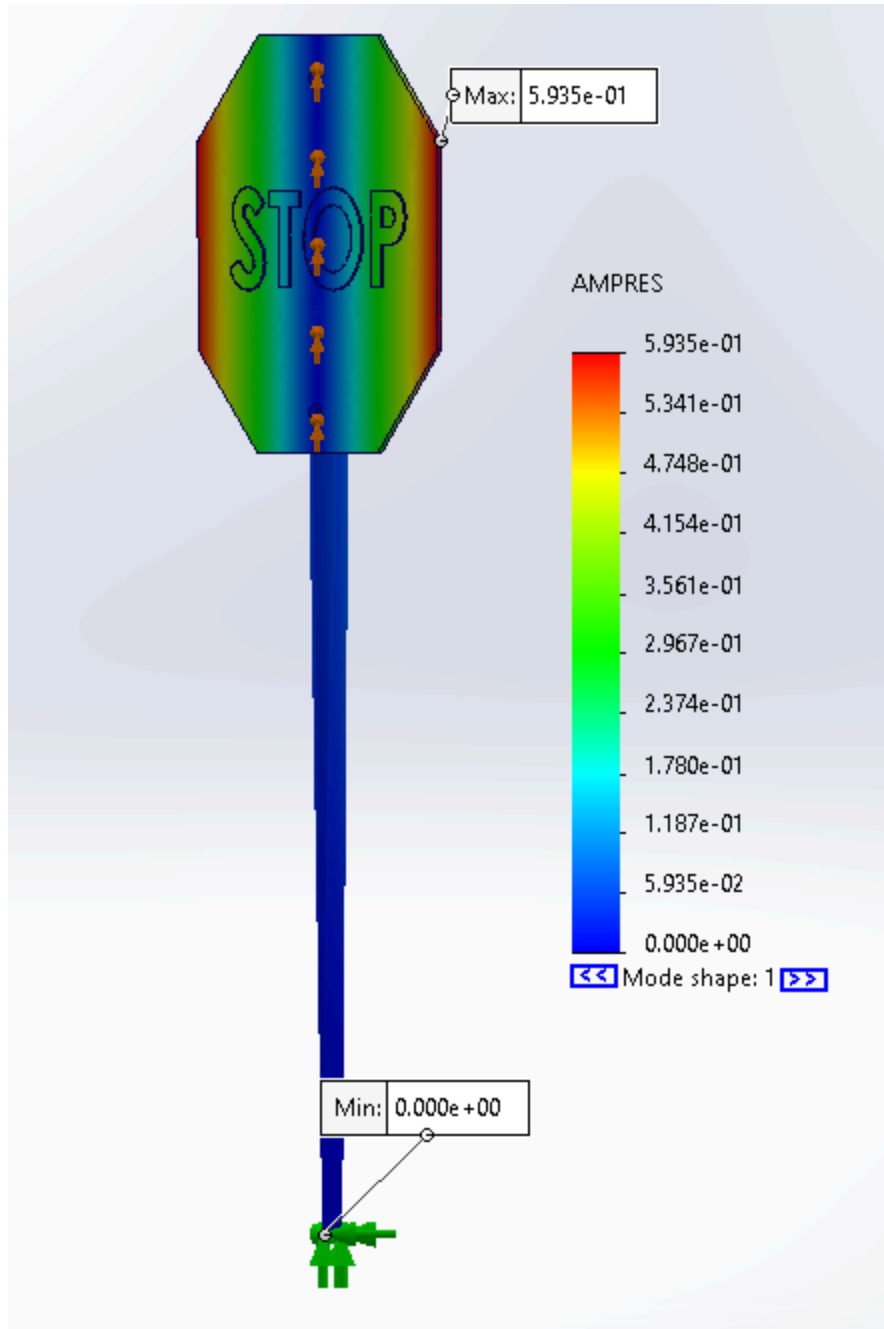


Figure 32. Angled Side View of Mode 1 from Simulation 3

Mode Shape 2: 6.9634 Hz (Second Bending Mode)

Mode Shape 2 shows a higher order bending pattern with one internal node along the post length. The displacement amplitudes remain highest at the top of the sign, but the curvature reverses direction at roughly the upper third of the post.

This mode is stiffer and therefore higher in frequency, with a natural frequency of 6.9634 Hz. This frequency is significantly higher than typical vortex-shedding frequencies for a post of similar diameter under hurricane-speed winds, making resonance with this mode unlikely. Still, its presence defines the dynamic envelope of the system.

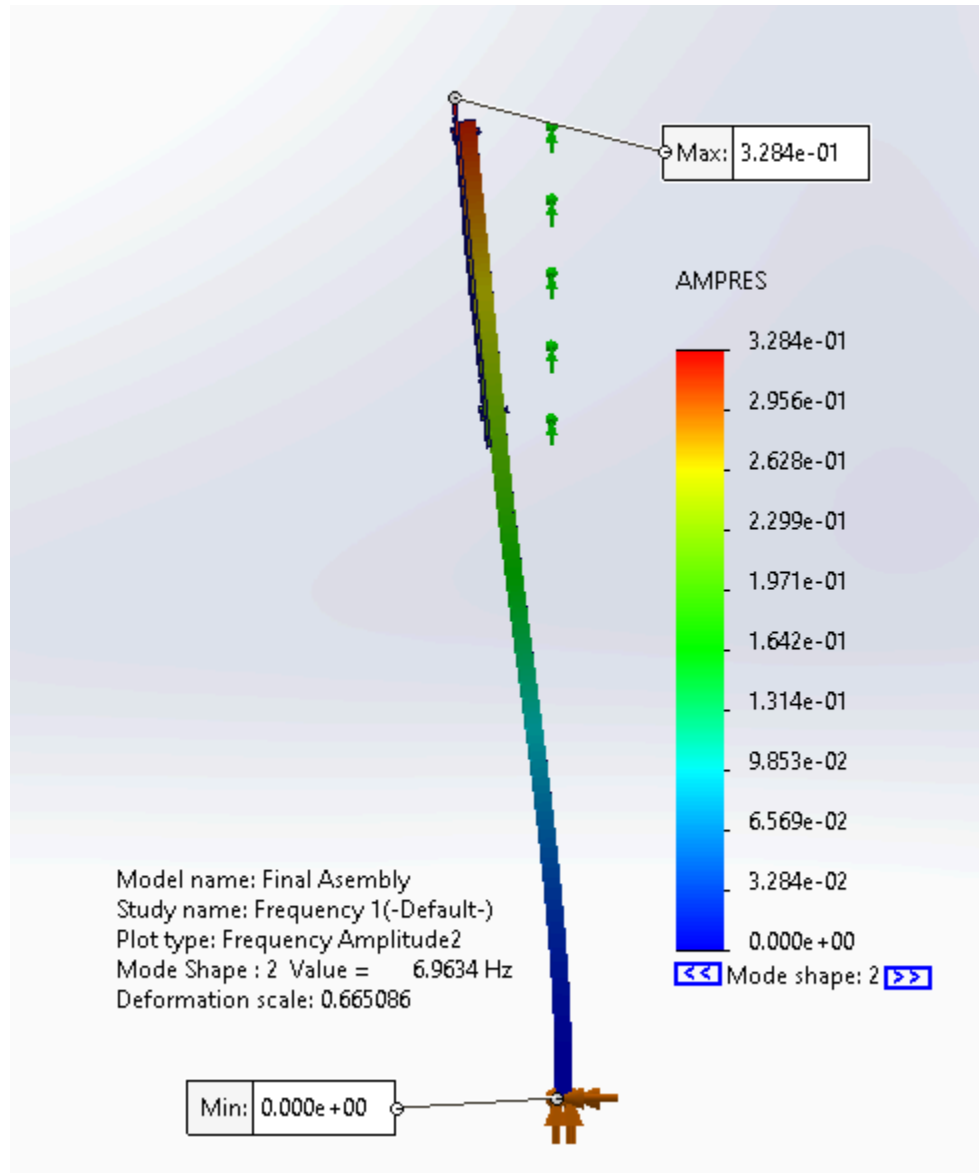


Figure 33. Mode 2 Sideview - Simulation 3

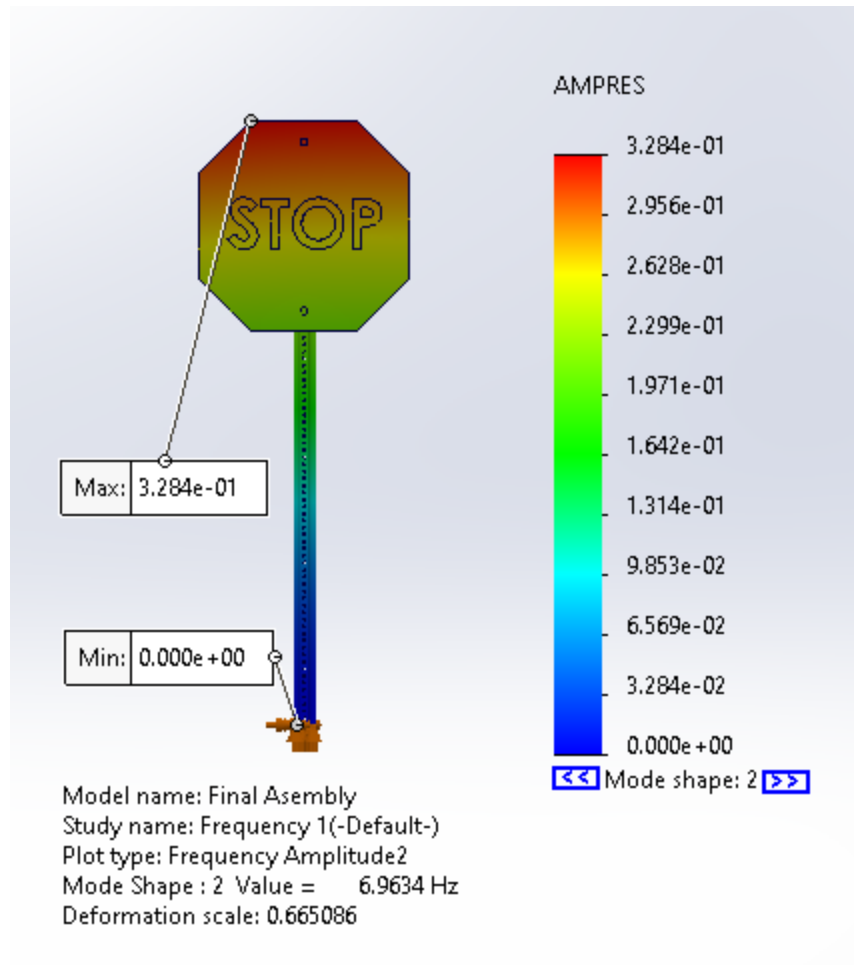


Figure 34. Mode 2 Frontview - Simulation 3

Mode Shape 3: 48.826 Hz (Higher-Order Local Mode)

Mode Shape 3 exhibits higher-frequency deformation dominated by localized curvature near the midsection of the post and the interface between the post and the sign. The maximum amplitude is located near the mid-height of the post, with minimal contribution at the fixed base.

With a natural frequency of 48.826 Hz, this mode is far above any wind-driven excitation likely to occur in real conditions. It represents a higher-order, stiffer vibrational behavior primarily governed by the post's geometry rather than the mass of the sign.

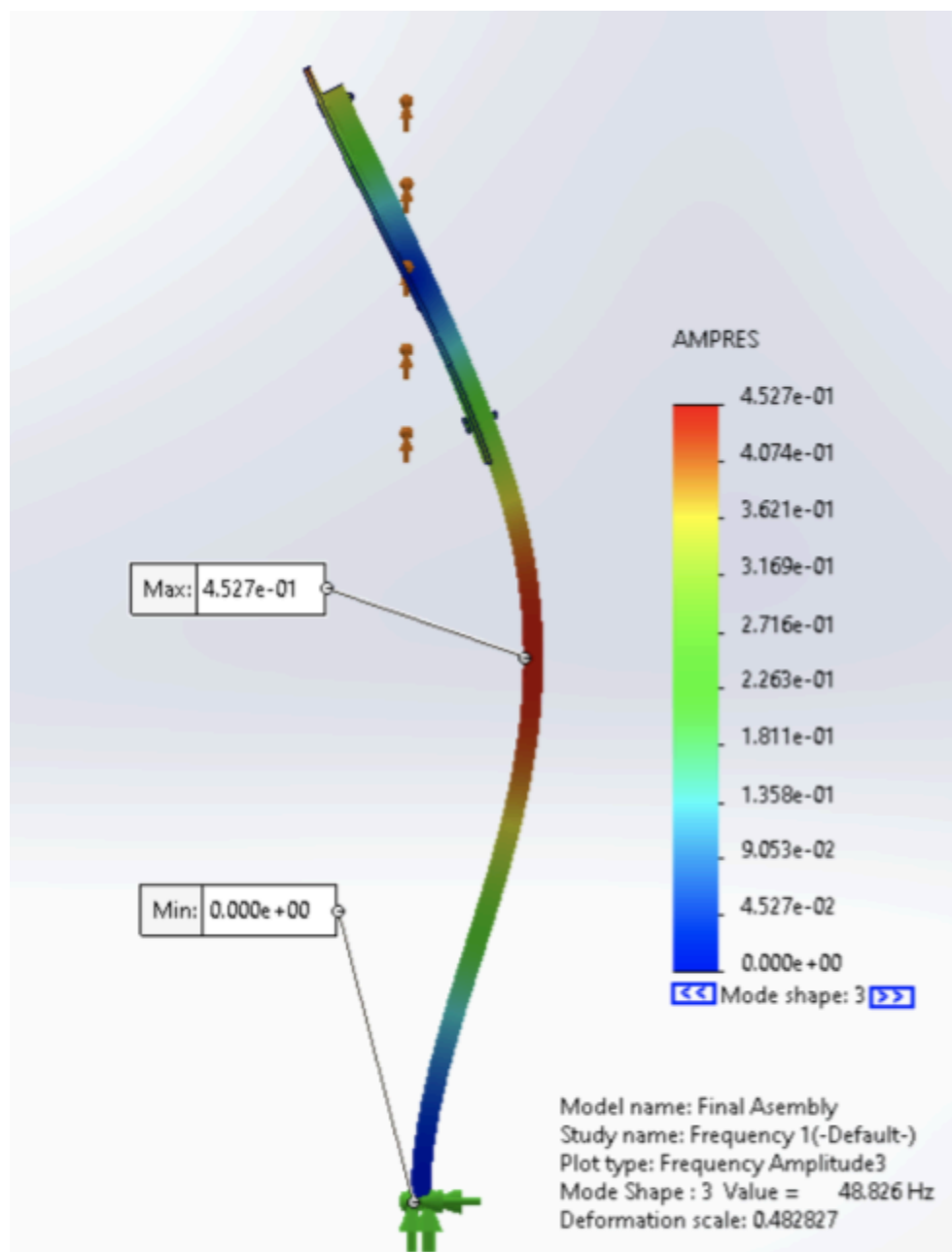


Figure 35. Mode 2 Side View - Simulation 3

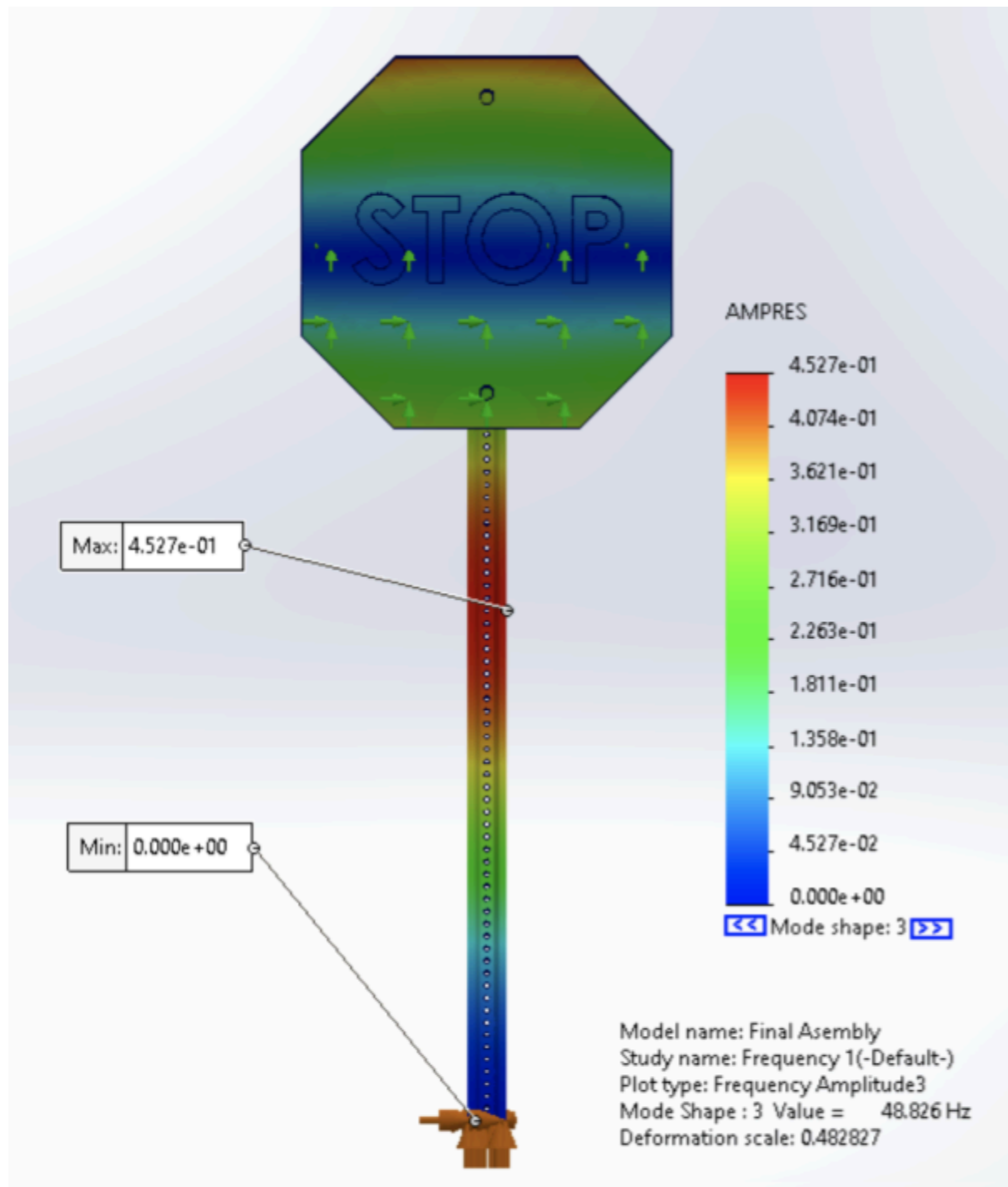


Figure 36. Mode 3 Front View - Simulation 3

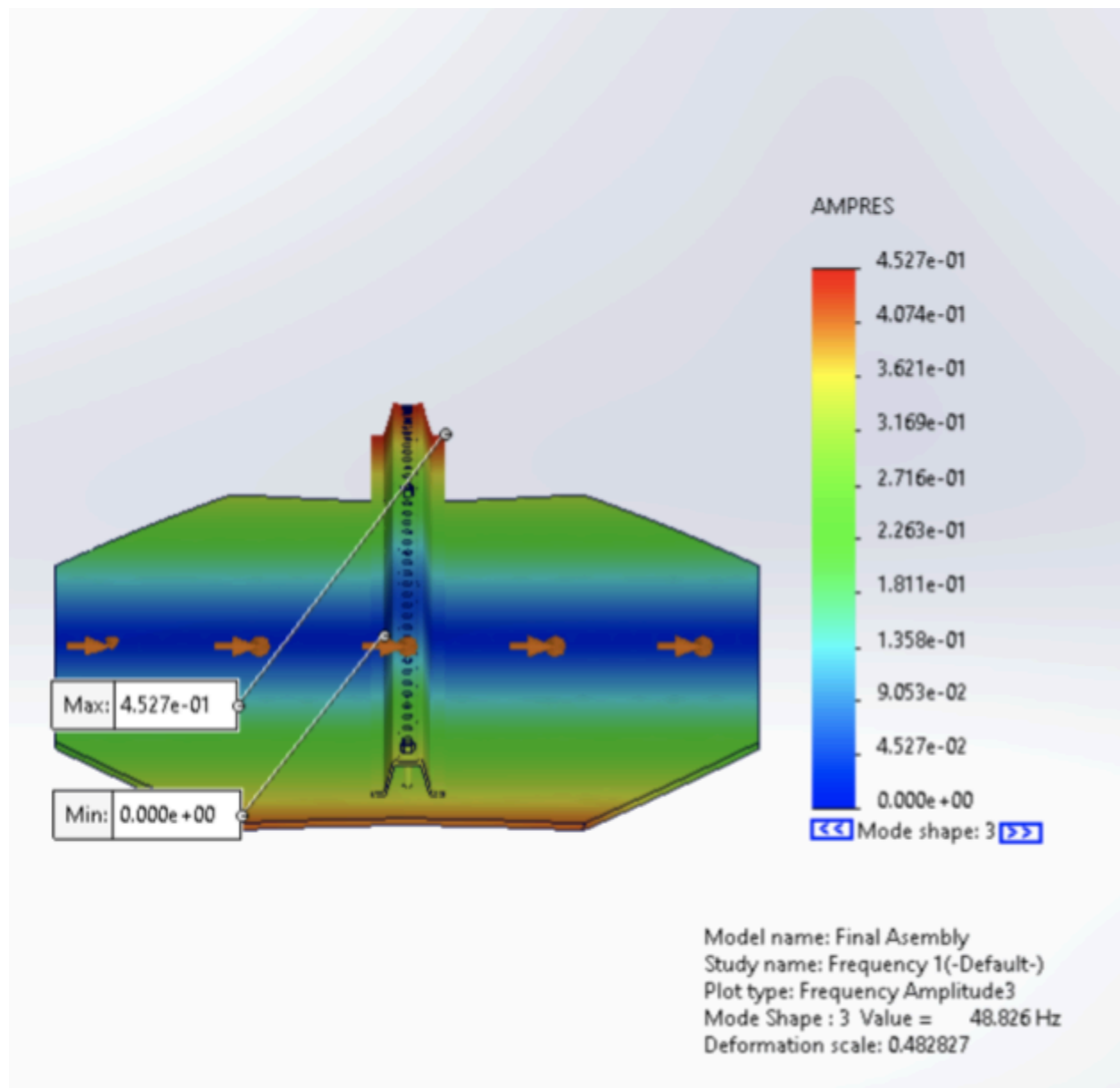


Figure 37. Mode 3 Back View of Plate - Simulation 3

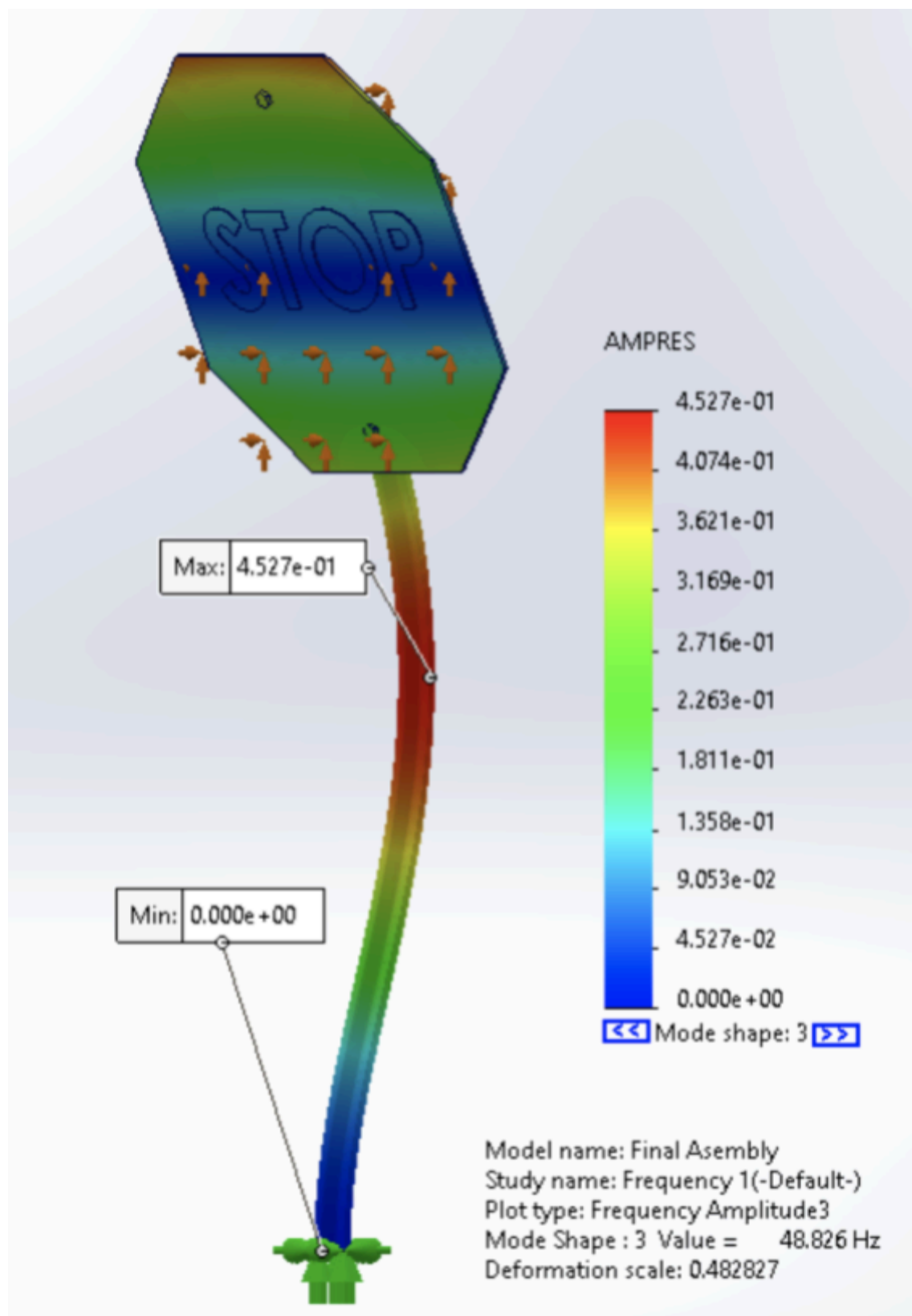


Figure 38. Mode 3 Side View - Simulation 3

Mode No.	Frequency(Rad/sec)	Frequency(Hertz)	Period(Seconds)
1	36.682	5.8381	0.17129
2	43.753	6.9634	0.14361
3	306.78	48.826	0.020481

Figure 39. Summary Table of Mode and Frequencies

Discussion:

Fluid Dynamics Simulations:

Drag dominated the loading, confirming that the sign behaves as a bluff body with significant pressure buildup on the windward face and complete flow separation at the edges.

The aerodynamic results show that the 30×30 inch stop sign behaves as a classic bluff body when exposed to hurricane-level winds. The pressure field is dominated by a strong stagnation region at the center of the windward face and full flow separation along all eight edges, producing a large, steady drag force of approximately 207 lbf. The goal plots confirm stable convergence, with drag remaining nearly constant and lift approaching zero exactly what is expected for a flat plate normal to the flow.

The wake region behind the sign forms a broad unsteady turbulence zone where vortex shedding occurs. Although this unsteadiness does not significantly affect the mean aerodynamic forces, it introduces fluctuating pressure that becomes relevant for dynamic structural loading. Overall, the fluid simulation provides a reliable and physically consistent pressure distribution for the subsequent FEA, capturing both the steady drag load and the characteristics of the separated, hurricane-speed flow around the sign.

FEA Static

The simulation results show that the STOP sign assembly experiences its highest structural demand at the base of the post, where both stress and strain reach their peak values under the applied wind load of 206.86 lbf. This behavior is expected for a cantilevered structure: the post must resist the entire bending moment generated by the drag force acting on the elevated sign plate. Because the plate is stiff relative to the post, it contributes little to deformation, which explains why the stress and strain across the octagonal sign remain minimal.

The maximum von Mises stress of 123,900 psi is far above the yield strength of common galvanized steel and aluminum alloys, meaning that the sign post would plastically deform or fail under such a wind load. The large displacement (≈ 5.7 in) at the top of the assembly reinforces this conclusion, showing that excessive bending would occur before reaching ultimate load capacity. The strain distribution also confirms that most deformation occurs in the lower section of the post, while the rest of the structure remains relatively rigid.

Overall, the analysis suggests that under extreme wind conditions equivalent to the hurricane-level drag forces used, the current geometry and materials of the post are insufficient to maintain structural integrity. The stop sign plate itself is not the limiting factor in durability; instead, the post requires redesign either through thicker material, stiffer cross-section, or reinforced mounting to meet high load conditions.

Calculations

$$1 \text{ N/mm}^2 = 145.038 \text{ psi}$$

$$E = 200,000 \text{ N/mm}^2 = 200,000 \times 145.038 = 2.90 \times 10^7 \text{ psi}$$

$$E = 2.90 \times 10^7$$

$$I = bh^3/12$$

$$b=h=2\text{in}$$

$$I = (2 \times 2^4)/12 = 16/12 = 4/3 = 1.33 \text{ in}^4$$

$$\delta = PL^3 / 3EI$$

$$P=206.86 \text{ lbf}$$

$$L=84 \text{ in} = 84 \text{ in}$$

$$E=2.90 \times 10^7$$

$$I=4/3 \text{ in}^4$$

$$L^3 = 84^3 = 592,704 \text{ in}^3$$

$$\delta = (206.86 \times 592,704) / (3 \times (2.90 \times 10^7) \times (4/3))$$

$$\delta = 1.06 \text{ inches}$$

To verify the structural behavior predicted by the FEA, a simplified hand calculation was performed by modeling the stop-sign post as a solid 2×2 inch cantilever beam with a fixed base and a lateral load applied at the free end. The applied load was the hurricane-induced drag force of 206.86 lbf obtained from the first simulation. Using the elastic modulus of galvanized steel and the moment of inertia for a 2×2 in square section, the classical cantilever beam formula was used to estimate tip deflection. For an 84-inch post length, the calculated displacement was approximately 1.06 inches. This value is significantly lower than the displacement predicted by the FEA because the hand calculation assumes a solid square post, which is much stiffer than the real U channel cross-section. This comparison demonstrates that the simplified beam model provides a conservative lower-bound estimate, while the FEA captures the reduced stiffness and higher flexibility of the actual post geometry, making it a more accurate representation of real world behavior.

These simplifications are valid because the goal of the hand calculation is not to perfectly replicate the geometry of the U-channel post, but to establish a baseline estimate that can be compared to the FEA. Treating the post as a solid 2×2 inch cantilever provides a stiffer, idealized version of the real structure, which ensures that the hand calculation produces a lower-bound deflection rather than an exaggerated one. This approach is common in verification work because it isolates the fundamental bending behavior without the added complexity of the true cross-section. If the FEA predicts greater deflection, and it does then the simplified model has successfully confirmed that the simulation is capturing real geometric flexibility rather than producing artificially high displacement.

To validate the aerodynamic loading, the drag force predicted by the flow simulation was compared to a hand calculation based on a flat-plate drag model at the same Reynolds number. Using an estimated drag coefficient for a bluff plate and the projected area of the 30×30 in sign, the analytical drag was computed and then compared against the integrated pressure forces from the CFD solution. The values showed reasonable agreement (on the order of $\sim 10\%$), indicating that the simulated pressure field is consistent with standard aerodynamic theory. For the structure, peak bending stress in the post was estimated using simple beam theory with the hand-calculated drag force applied at the sign centroid. These analytical stresses were then compared with the FEA results at corresponding locations, again showing similar magnitudes and trends. In the full report, these comparisons are summarized in tables and plots to clearly show the agreement in drag force, bending moment, and maximum stress.

Small discrepancies between the simulation and analytical results are expected and were observed. The CFD model predicts a more detailed pressure distribution (rather than a uniform load), leading to slightly different moment and stress profiles than the simplified flat-plate assumption. Likewise, the FEA model includes geometric and stiffness details (bolt locations, U channel shape, local stress concentrations) that are not captured in simple beam equations, which treat the post as a uniform prismatic member.

Sources of error in the model include idealized boundary conditions (perfectly fixed post base, no soil flexibility), assumptions of homogeneous and isotropic material properties, mesh resolution near sharp edges and in the wake region, and the choice of turbulence model. Additionally, the mapping of CFD pressures onto the structural mesh introduces interpolation error, and the use of steady or simplified loading profiles in the structural analysis neglects some of the full variability of real hurricane gusts. These factors were mitigated through mesh refinement studies and sensitivity checks, but they remain important limitations when interpreting the exact numerical values of the results.

Modal Analysis: Vibrations FEA

The modal analysis of the stop-sign assembly provides important context for interpreting its response to hurricane-level wind loading. The first three natural frequencies were found to be approximately 0.59 Hz, 6.96 Hz, and 48.83 Hz, with corresponding mode shapes dominated by global bending of the post and sign. These results confirm that the structure behaves primarily as a tall, flexible cantilever with a relatively heavy tip mass (the sign plate).

The fundamental mode at 0.59 Hz is a long period, global bending mode in which the entire post deflects laterally and the sign moves as a rigid body at the tip. This low frequency is consistent with the large slenderness ratio of the post and the fixed base boundary condition. In real wind conditions, excitation at this frequency is more likely to come from large scale gusting or overall buffeting of the flow than from classic vortex shedding, which typically occurs at higher frequencies for a narrow post. However, because dynamic amplification is largest near the fundamental frequency, this mode defines the most critical range for potential serviceability issues such as noticeable swaying or oscillation of the sign.

The second mode at 6.96 Hz shows a higher-order bending shape with an internal node along the post. Its much higher frequency implies that considerably faster, more localized wind fluctuations would be required to excite it. For the hurricane wind scenario considered in this study, the flow is unlikely to contain persistent energy at this frequency, so significant resonance with Mode 2 is improbable. Nevertheless, this mode is relevant for understanding the global dynamic stiffness of the system and would become more important under impact or short-duration loading, such as vehicle strike or sudden release of tension.

The third mode at 48.83 Hz is characterized by localized deformation concentrated in the post and connection region, with limited contribution from rigid body motion of the sign. The very high frequency and relatively small deformation wavelengths indicate that this mode is governed by the local stiffness of the post cross-section and the connection details. Under wind loading, which typically has little energy at such high frequencies, this mode is not expected to

be excited to a meaningful amplitude. Its main value lies in confirming that no low-frequency local modes exist that might couple strongly with aerodynamic phenomena.

When the modal results are considered alongside the static analysis, a consistent picture emerges. The static study showed that stresses and strains are strongly concentrated at the base of the post, and that maximum displacements occur at the top of the sign. The fundamental mode shape mirrors this deformation pattern, indicating that under dynamic loading the same regions will remain critical. Because the higher modes are stiffer and involve smaller overall deflections, they are less likely to govern either strength or serviceability for the wind speeds considered.

Overall, the modal analysis suggests that the stop sign structure is most vulnerable to low-frequency, large-amplitude excitation near 0.6 Hz, corresponding to global swaying of the post. For the hurricane case examined here, the wind environment is not expected to drive strong resonance at the second and third modes, and the dynamic behavior is therefore dominated by the first bending mode. This supports the use of quasistatic design checks based on the derived drag load, with additional consideration of fatigue and serviceability only if the sign were installed in a location known to experience persistent low-frequency gusting or other periodic excitation.

VI. Conclusion

This project analyzed the aerodynamic and structural behavior of a standard 30×30 in MUTCD stop sign exposed to Category 2 hurricane winds using SolidWorks. The flow simulation showed that the sign acts as a bluff body, developing a strong stagnation region on the front face and full flow separation along its edges. This produced a steady drag force of about 207 lbf and essentially zero lift, confirming that hurricane loading on the sign is dominated by drag.

The static structural analysis demonstrated that the U channel post is the critical structural element. Under the hurricane level drag load, stresses at the fixed base of the post exceeded typical yield strength, and the top of the sign deflected several inches. The aluminum plate carried little stress compared to the post, indicating that failure or permanent deformation would initiate in the steel support rather than the sign face.

The modal analysis reinforced this conclusion: the fundamental natural frequency (≈ 0.59 Hz) corresponds to a large-scale bending mode of the post, meaning that any dynamic motion under wind will primarily involve global swaying rather than localized vibration. Higher modes occurred at much higher frequencies and are unlikely to be excited by hurricane wind fluctuations.

Overall, the results indicate that a standard U channel-mounted 30 × 30 stop sign is structurally vulnerable under upper range Category 2 winds. The post, not the sign plate requires reinforcement or redesign to ensure performance in extreme weather. Strengthening the post, reducing unsupported height, or using a stiffer cross-section would significantly improve resilience in hurricane-prone areas.

VII. Appendix

Equations used:

[1] Lift Force Y:

$$F_L = \frac{1}{2} \rho V^2 A C_L$$

[2] Coefficient of Lift :

$$C_L = \frac{L}{A \cdot q}$$

where $q = \frac{\rho V^2}{2}$ ρ is air density, and V is airspeed

[3] Drag Force X:

$$F_D = \frac{1}{2} \rho V^2 A C_D$$

[4] Coefficient of Drag:

$$C_D = \frac{F_D}{A \cdot q}$$

[5] Force:

$$F = ma$$

[6] Elastic Modulus Conversion:

$$1 \text{ N/mm}^2 = 145.038 \text{ psi}$$

[7] Area Moment of Inertia:

$$I = bh^3 / 12$$

[8] Cantilever Beam Tip Deflection

$$\delta = PL^3 / 3EI$$

[9] Beam Length Cubed

$$L^3$$

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